Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems

MSc Research Internship Report

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April – May 2026

Acknowledgments

I warmly thank Martin J. Gander and Bastien Chaudet-Dumas for their guidance, availability, and support throughout this research internship. This project gave me the opportunity to study a stimulating problem and to use the tools in numerical analysis and partial differential equations developed during my MSc year.

Introduction

The Lotka–Volterra system is a classical prey–predator model. Volterra introduced it to describe variations in interacting biological populations [21], and it is now a standard model in mathematical biology [14]. In this report, we study the classical time-dependent equations and their spatial extension with diffusion, using fish populations in a lake as the main interpretation.

The first part presents numerical simulations of the spatial Lotka–Volterra system. The model consists of reaction–diffusion equations with homogeneous Neumann conditions, so the normal population flux satisfies $J_i \cdot n = 0$ on the boundary. We use finite differences, following [8, 15, 16]; the general reaction–diffusion setting is also presented in [10].

The second part introduces an optimal control problem in which fishing acts on the prey population. The cost has the form $J = J_{\text{track}} + J_{\text{effort}} - J_{\text{yield}}$: it measures the distance to target profiles, penalizes the fishing effort, and rewards the harvested quantity. Using optimal control for evolution equations and PDE-constrained problems [12, 20], together with Pontryagin's minimum principle [19], we derive the adjoint equations and characterize the optimal control.

We then discretize the optimality system with a forward–backward sweep algorithm and compare the results with projected-gradient and L-BFGS-B methods. We also study stationary profiles and prove a conditional convergence result for the fully discrete iteration, using the contraction ideas of McAsey, Mou and Han [13].

Contents

1 The Lotka–Volterra System in Time and Space . . . 1
- 1.1 Numerical Approximation in One Space Dimension . . . 1
- 1.2 Numerical Approximation in Two Space Dimensions . . . 4

2 Control Problems . . . 6
- 2.1 Model Without a Spatial Variable and Constant Controls . . . 6
- 2.2 Space-time Optimal Control Problem . . . 8
- 2.3 Stationary Profiles . . . 17
- 2.4 Numerical Computation of the Optimal Control . . . 19
- 2.5 Fully Discrete Conditional Convergence of the Forward–Backward Sweep Method . . . 24

1 The Lotka–Volterra System in Time and Space

1.1 Numerical Approximation in One Space Dimension

We start with the classical Lotka–Volterra model [14, 21]; numerical schemes and qualitative properties are discussed in [3, 7]. Let $v_1(t)$ be the predator density and $v_2(t)$ the prey density at time t. Without spatial dependence, the state is $v(t) = (v_1(t), v_2(t)) \in \mathbf{R}_+^2$. We assume $v_1, v_2 \in \mathscr{C}^1([0,T];\mathbf{R}_+)$, so both densities are differentiable and nonnegative. They solve

$$\begin{cases} \dot{v}_1(t) = v_1(t)\big(-m + \alpha v_2(t)\big), \\ \dot{v}_2(t) = v_2(t)\big(r - \beta v_1(t)\big), \end{cases} \tag{1}$$

The parameters satisfy $m, r, \alpha, \beta > 0$. Here m is the natural predator mortality rate and r is the prey growth rate without predators. The coefficient α measures the positive effect of prey on predator growth, while β measures the loss of prey through predation. To include movement, let $\Omega \subseteq \mathbf{R}^d$ be a bounded habitat and let $x \in \Omega$ denote position. The unknown densities become

$$v_1, v_2 : \Omega \times [0,T] \longrightarrow \mathbf{R}_+.$$

For the pointwise calculations below, we assume enough regularity; for example,

$$v_i \in \mathscr{C}^1([0,T];\mathscr{C}^0(\overline{\Omega})) \cap \mathscr{C}^0([0,T];\mathscr{C}^2(\overline{\Omega})), \qquad i \in \{1,2\},$$

Under this regularity assumption, every derivative is defined pointwise and the spatial Lotka–Volterra system is

$$\begin{cases} \partial_t v_1(x,t) = v_1(x,t)\big(-m + \alpha v_2(x,t)\big) + \mu_1 \Delta v_1(x,t), & \text{in } \Omega \times (0,T), \\ \partial_t v_2(x,t) = v_2(x,t)\big(r - \beta v_1(x,t)\big) + \mu_2 \Delta v_2(x,t), & \text{in } \Omega \times (0,T), \\ \quad \dfrac{\partial v_1}{\partial n} = \dfrac{\partial v_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0,T), \\ \quad v_1(\cdot,0) = v_1^0, \qquad v_2(\cdot,0) = v_2^0, & \text{in } \Omega. \end{cases} \tag{2}$$

Here $x \in \mathbf{R}^d$ is the spatial variable, Δ is the Laplace operator, and $\mu_1, \mu_2 > 0$ are the diffusion coefficients. The term $\mu_i \Delta v_i$ models random movement of species i. At the macroscopic scale, Fick's law relates the population flux to the density gradient: $J_i = -\mu_i \nabla v_i$. Thus individuals move, on average, from high-density regions toward low-density regions. If $F_i(v_1, v_2)$ denotes the local Lotka–Volterra reaction, conservation on a small volume gives

$$\partial_t v_i + \nabla \cdot J_i = \text{reaction terms}.$$

After substituting $J_i = -\mu_i \nabla v_i$, this becomes

$$\partial_t v_i - \mu_i \Delta v_i = \text{reaction terms}.$$

The initial densities satisfy $v_1^0, v_2^0 : \Omega \to \mathbf{R}_+$. We impose homogeneous Neumann conditions [6, 10], equivalently $\partial_n v_i = 0$ or $J_i \cdot n = 0$ on $\partial\Omega$. Thus the habitat is closed and no population crosses its boundary. We first take $d = 1$ and $\Omega = (0,1)$, so

$$v_1, v_2 : (0,1) \times [0,T] \longrightarrow \mathbf{R}_+.$$

The Laplacians then reduce to

$$\Delta v_1(x,t) = \partial_x^2 v_1(x,t), \qquad \Delta v_2(x,t) = \partial_x^2 v_2(x,t),$$

and system (2) becomes

$$\begin{cases} \partial_t v_1(x,t) = v_1(x,t)\big(-m + \alpha v_2(x,t)\big) + \mu_1 \partial_x^2 v_1(x,t), & \text{in } (0,1) \times (0,T), \\ \partial_t v_2(x,t) = v_2(x,t)\big(r - \beta v_1(x,t)\big) + \mu_2 \partial_x^2 v_2(x,t), & \text{in } (0,1) \times (0,T), \\ \partial_x v_1(0,t) = \partial_x v_1(1,t) = 0, & t \in (0,T), \\ \partial_x v_2(0,t) = \partial_x v_2(1,t) = 0, & t \in (0,T), \\ \quad v_1(\cdot,0) = v_1^0, \qquad v_2(\cdot,0) = v_2^0, & \text{in } (0,1). \end{cases} \tag{3}$$

To obtain a finite-dimensional approximation, discretize $[0,1]$ by the uniform grid $0 = x_0 < x_1 < \cdots < x_N = 1$. Its spacing is $h = x_{i+1} - x_i = 1/N$, and hence $x_i = ih$ for every $i = 0, \dots, N$. We represent the two densities by their nodal values

$$v_{1,i}(t) \approx v_1(x_i,t), \qquad v_{2,i}(t) \approx v_2(x_i,t).$$

The semi-discrete state is therefore the pair of time-dependent vectors

$$V_1(t) = (v_{1,0}(t), \ldots, v_{1,N}(t)) \in \mathbf{R}_+^{N+1}, \qquad V_2(t) = (v_{2,0}(t), \ldots, v_{2,N}(t)) \in \mathbf{R}_+^{N+1}.$$

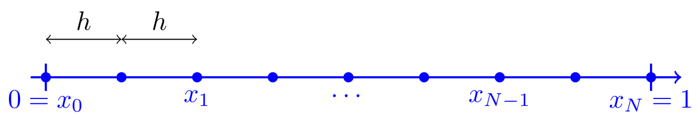

Figure 1: Uniform discretization of the spatial interval $[0, 1]$.

At an interior node x_i, $i = 1, \ldots, N-1$, Taylor expansion gives the centered finite-difference approximation of order two [8, 16]:

$$\partial_x^2 v_1(x_i, t) \approx \frac{v_{1,i+1}(t) - 2v_{1,i}(t) + v_{1,i-1}(t)}{h^2}, \qquad \partial_x^2 v_2(x_i, t) \approx \frac{v_{2,i+1}(t) - 2v_{2,i}(t) + v_{2,i-1}(t)}{h^2}.$$

The interior semi-discrete equations obtained from system (3) are therefore

$$\begin{cases} \partial_t v_{1,i}(t) = v_{1,i}(t)\big(\alpha v_{2,i}(t) - m\big) + \mu_1 \dfrac{v_{1,i+1}(t) - 2v_{1,i}(t) + v_{1,i-1}(t)}{h^2}, & i = 1, \ldots, N-1, \\ \partial_t v_{2,i}(t) = v_{2,i}(t)\big(r - \beta v_{1,i}(t)\big) + \mu_2 \dfrac{v_{2,i+1}(t) - 2v_{2,i}(t) + v_{2,i-1}(t)}{h^2}, & i = 1, \ldots, N-1, \\ \quad v_{1,-1} = v_{1,1}, \qquad v_{1,N+1} = v_{1,N-1}, \\ \quad v_{2,-1} = v_{2,1}, \qquad v_{2,N+1} = v_{2,N-1}. \end{cases} \tag{4}$$

At x_0 and x_N, centered differences and ghost points impose the Neumann conditions:

$$\frac{v_{k,1}(t) - v_{k,-1}(t)}{2h} = 0, \qquad \frac{v_{k,N+1}(t) - v_{k,N-1}(t)}{2h} = 0, \qquad k = 1, 2.$$

Hence

$$v_{k,-1}(t) = v_{k,1}(t), \qquad v_{k,N+1}(t) = v_{k,N-1}(t), \qquad k = 1, 2.$$

These ghost values are exactly the discrete form of the zero normal derivative. Substitution into the centered second difference gives the endpoint Laplacians

$$(\delta_{xx} v_k)_0 = \frac{2\big(v_{k,1} - v_{k,0}\big)}{h^2}, \qquad (\delta_{xx} v_k)_N = \frac{2\big(v_{k,N-1} - v_{k,N}\big)}{h^2}, \qquad k = 1, 2.$$

The boundary construction is detailed in [6, 15]. The Python code is available in the GitHub repository [2], and the corresponding simulation video is available here. We choose the localized Gaussian initial densities

$$v_1(x, 0) = 1.5 \exp\left(-\frac{(x - 0.3)^2}{2 \cdot 0.12^2}\right) + 0.02, \qquad v_2(x, 0) = 1.4 \exp\left(-\frac{(x - 0.7)^2}{2 \cdot 0.08^2}\right) + 0.02.$$

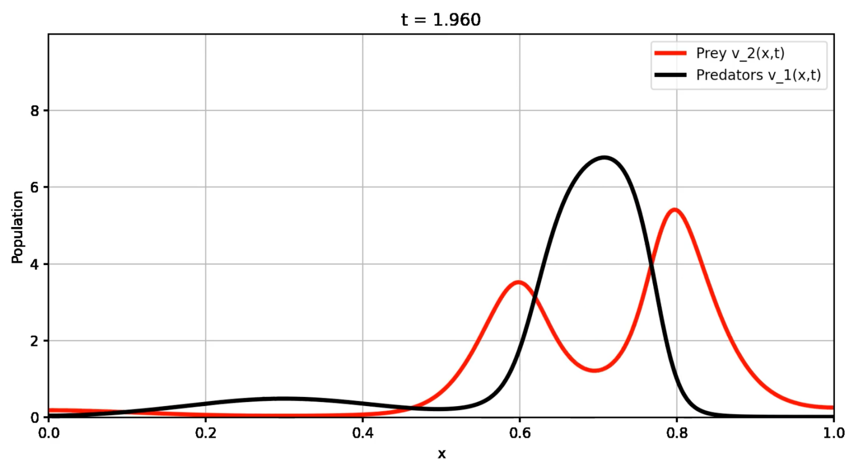

Figure 2: Predator and prey profiles at $t = 1.960$ with Gaussian initial data and homogeneous Neumann boundary conditions.

For the computation shown in Figure 2, the spatial finite-difference method produces an ODE in $\mathbf{R}^{2(N+1)}$. We integrate this ODE by explicit Euler on the uniform time grid

$$0 = t^0 < t^1 < \cdots < t^{N_t} = T, \qquad \Delta t = t^{n+1} - t^n,$$

and write

$$v_{k,i}^n \approx v_k(x_i, t^n), \qquad k = 1, 2.$$

In the code, the notation is reversed relative to the mathematical ordering: $u_i^n = v_{2,i}^n$ denotes the prey density, while $v_i^n = v_{1,i}^n$ denotes the predator density. With this convention, the time updates are

$$u_i^{n+1} = u_i^n + \Delta t \left[u_i^n(b - \beta v_i^n) + \nu_u \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{h^2} \right],$$

and

$$v_i^{n+1} = v_i^n + \Delta t \left[v_i^n(\alpha u_i^n - a) + \nu_v \frac{v_{i+1}^n - 2v_i^n + v_{i-1}^n}{h^2} \right],$$

for $i = 1, \ldots, N-1$. In the implementation, $a = m$, $b = r$, $u = v_2$, $v = v_1$, $\nu_u = \mu_2$, and $\nu_v = \mu_1$. The same Euler formula is used at $i = 0$ and $i = N$, with the modified discrete Laplacian obtained from the ghost points. At interior nodes, the spatial stencil has order two, and explicit Euler has order one in time. Thus the formal error is $O(\Delta t + h^2)$ for a smooth solution as long as the positivity projection does not alter the update. Stability imposes an additional relation between h and Δt. If only diffusion is retained, a sufficient condition is

$$\Delta t \le \frac{h^2}{2 \max(\nu_u, \nu_v)},$$

The reaction terms can impose a smaller value of Δt. The implementation [2] also applies the pointwise projection $z \mapsto \max(z, 0)$ after each time step to remove negative values created by the explicit scheme; see [3, 7]. The order stated above applies when this projection is inactive.

To compare boundary effects, we repeat the one-dimensional experiment with homogeneous Dirichlet conditions,

$$v_1(0, t) = v_1(1, t) = 0, \qquad v_2(0, t) = v_2(1, t) = 0.$$

The discrete boundary values are then fixed by

$$v_{1,0}^n = v_{1,N}^n = 0, \qquad v_{2,0}^n = v_{2,N}^n = 0, \qquad n = 0, \ldots, N_t.$$

Figure 3 shows the resulting profiles.

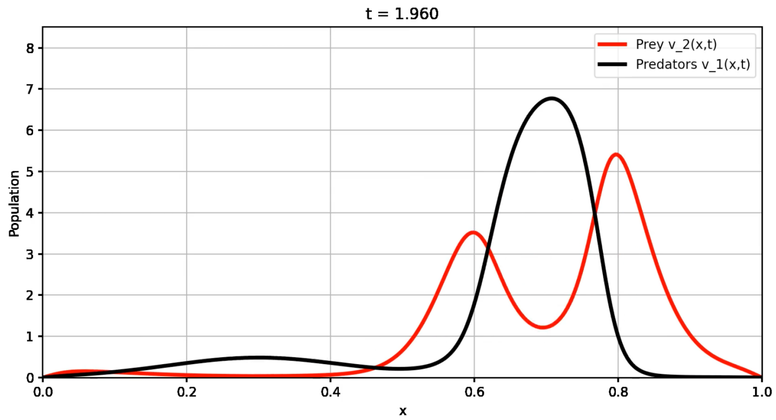

Figure 3: Predator and prey profiles at $t = 1.960$ with Gaussian initial data and homogeneous Dirichlet boundary conditions.

Remark. Gaussian data describe populations initially concentrated in small regions; diffusion then spreads each profile. Constant data, sinusoidal perturbations, step functions, and sums of Gaussians produce different transients. Neumann conditions model a closed habitat, while Dirichlet conditions model an absorbing or hostile boundary.

1.2 Numerical Approximation in Two Space Dimensions

The same finite-difference construction extends to $d = 2$. We take $(x, y) \in \mathbf{R}^2$ and use the framework of [8, 16]. The densities are now

$$v_1, v_2 : (0, 1)^2 \times [0, T] \longrightarrow \mathbf{R}_+.$$

For $w = w(x, y, t)$, the two-dimensional Laplacian is $\Delta w = \partial_x^2 w + \partial_y^2 w$. Hence

$$\Delta v_1(x, y, t) = \partial_x^2 v_1(x, y, t) + \partial_y^2 v_1(x, y, t), \qquad \Delta v_2(x, y, t) = \partial_x^2 v_2(x, y, t) + \partial_y^2 v_2(x, y, t).$$

System (2) becomes

$$\begin{cases} \partial_t v_1(x, y, t) = v_1(x, y, t)\big(\alpha v_2(x, y, t) - m\big) + \mu_1 \Delta v_1(x, y, t), & \text{in } (0, 1)^2 \times (0, T), \\ \partial_t v_2(x, y, t) = v_2(x, y, t)\big(r - \beta v_1(x, y, t)\big) + \mu_2 \Delta v_2(x, y, t), & \text{in } (0, 1)^2 \times (0, T), \\ \qquad \dfrac{\partial v_1}{\partial n} = \dfrac{\partial v_2}{\partial n} = 0, & \text{on } \partial(0, 1)^2 \times (0, T), \\ \quad v_1(\cdot, \cdot, 0) = v_1^0, \qquad v_2(\cdot, \cdot, 0) = v_2^0, & \text{in } (0, 1)^2. \end{cases}$$

We discretize $[0, 1]^2$ with the uniform tensor grid

$$0 = x_0 < \cdots < x_{N_x} = 1, \qquad 0 = y_0 < \cdots < y_{N_y} = 1,$$

where $h_x = N_x^{-1}$, $h_y = N_y^{-1}$, $x_i = ih_x$, and $y_j = jh_y$. The nodal values satisfy

$$v_{1,i,j}(t) \approx v_1(x_i, y_j, t), \qquad v_{2,i,j}(t) \approx v_2(x_i, y_j, t).$$

The semi-discrete state consists of the arrays

$$V_1(t) = \big(v_{1,i,j}(t)\big)_{0 \le i \le N_x,\, 0 \le j \le N_y}, \qquad V_2(t) = \big(v_{2,i,j}(t)\big)_{0 \le i \le N_x,\, 0 \le j \le N_y},$$

so $(V_1(t), V_2(t)) \in \big(\mathbf{R}_+^{(N_x+1)\times(N_y+1)}\big)^2$.

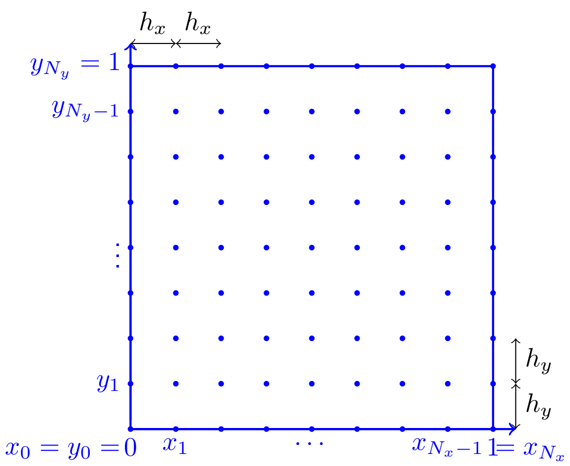

Figure 4: Uniform discretization of the spatial domain $[0, 1]^2$.

At an interior node (x_i, y_j), centered second differences give [8, 15, 16]

$$\partial_x^2 v_1(x_i, y_j, t) \approx \frac{v_{1,i+1,j}(t) - 2v_{1,i,j}(t) + v_{1,i-1,j}(t)}{h_x^2}, \qquad \partial_y^2 v_1(x_i, y_j, t) \approx \frac{v_{1,i,j+1}(t) - 2v_{1,i,j}(t) + v_{1,i,j-1}(t)}{h_y^2},$$

and therefore

$$\Delta v_1(x_i, y_j, t) \approx \frac{v_{1,i+1,j}(t) - 2v_{1,i,j}(t) + v_{1,i-1,j}(t)}{h_x^2} + \frac{v_{1,i,j+1}(t) - 2v_{1,i,j}(t) + v_{1,i,j-1}(t)}{h_y^2}.$$

The same five-point stencil for v_2 gives

$$\Delta v_2(x_i, y_j, t) \approx \frac{v_{2,i+1,j}(t) - 2v_{2,i,j}(t) + v_{2,i-1,j}(t)}{h_x^2} + \frac{v_{2,i,j+1}(t) - 2v_{2,i,j}(t) + v_{2,i,j-1}(t)}{h_y^2}.$$

The interior semi-discrete system is

$$\begin{cases} \partial_t v_{1,i,j} = v_{1,i,j}\big(\alpha v_{2,i,j} - m\big) + \mu_1 \left(\dfrac{v_{1,i+1,j} - 2v_{1,i,j} + v_{1,i-1,j}}{h_x^2} + \dfrac{v_{1,i,j+1} - 2v_{1,i,j} + v_{1,i,j-1}}{h_y^2} \right), & \substack{i=1,\dots,N_x-1, \\ j=1,\dots,N_y-1,} \\ \partial_t v_{2,i,j} = v_{2,i,j}\big(r - \beta v_{1,i,j}\big) + \mu_2 \left(\dfrac{v_{2,i+1,j} - 2v_{2,i,j} + v_{2,i-1,j}}{h_x^2} + \dfrac{v_{2,i,j+1} - 2v_{2,i,j} + v_{2,i,j-1}}{h_y^2} \right), & \substack{i=1,\dots,N_x-1, \\ j=1,\dots,N_y-1,} \\ \dfrac{\partial v_1}{\partial n} = \dfrac{\partial v_2}{\partial n} = 0, \qquad \text{on the boundary nodes.} \end{cases}$$

The boundary stencil depends on the boundary condition. For homogeneous Neumann conditions, we use the same reflected ghost-point construction as in (4). If $w_{i,j}$ denotes either v_1 or v_2, then $\partial_n w = 0$ is approximated by centered first differences. This gives

$$w_{-1,j} = w_{1,j}, \qquad w_{N_x+1,j} = w_{N_x-1,j},$$

on the vertical sides, and

$$w_{i,-1} = w_{i,1}, \qquad w_{i,N_y+1} = w_{i,N_y-1},$$

on the horizontal sides. Inserting the reflected values into the centered five-point stencil gives the boundary version of Δ_h. For example, at a node on the left side, $i = 0$ and $1 \le j \le N_y - 1$, we obtain

$$(\Delta_h w)_{0,j} = \frac{2(w_{1,j} - w_{0,j})}{h_x^2} + \frac{w_{0,j+1} - 2w_{0,j} + w_{0,j-1}}{h_y^2}.$$

The other three sides are obtained in the same way.

For the numerical experiment, we integrate the semi-discrete system with explicit Euler on the time grid

$$0 = t^0 < t^1 < \cdots < t^{N_t} = T, \qquad \Delta t = t^{n+1} - t^n,$$

and denote

$$u_{i,j}^n \approx v_2(x_i, y_j, t^n), \qquad v_{i,j}^n \approx v_1(x_i, y_j, t^n).$$

As in one dimension, $u_{i,j}^n = v_2(x_i, y_j, t^n)$ denotes prey and $v_{i,j}^n = v_1(x_i, y_j, t^n)$ denotes predators. The updates are

$$u_{i,j}^{n+1} = u_{i,j}^n + \Delta t \left[u_{i,j}^n(b - \beta v_{i,j}^n) + \nu_u \left(\frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{h_x^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{h_y^2} \right) \right],$$

and

$$v_{i,j}^{n+1} = v_{i,j}^n + \Delta t \left[v_{i,j}^n(\alpha u_{i,j}^n - a) + \nu_v \left(\frac{v_{i+1,j}^n - 2v_{i,j}^n + v_{i-1,j}^n}{h_x^2} + \frac{v_{i,j+1}^n - 2v_{i,j}^n + v_{i,j-1}^n}{h_y^2} \right) \right],$$

at the interior nodes. The code uses the notation $a = m$, $b = r$, $u = v_2$, $v = v_1$, $\nu_u = \mu_2$, and $\nu_v = \mu_1$. At boundary nodes, the same Euler formula is used with the reflected ghost values associated with $\partial_n v_i = 0$. The temporal approximation is first order and the five-point Laplacian is second order in each spatial direction. Thus, when the positivity projection is inactive and the exact solution is smooth, the formal error has the form $O(\Delta t + h_x^2 + h_y^2)$. Stability still requires a sufficiently small step. For the diffusion part alone, a sufficient condition is

$$\Delta t \, \max(\nu_u, \nu_v) \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) \le \frac{1}{2},$$

The reactions may require a smaller time step. After every step, the implementation [2] projects both densities onto $\mathbf{R}_+$. The stated order applies when the projection does not modify the solution.

The Python implementation is used to visualize the two-dimensional dynamics [2]; a video is available here. The initial densities are localized Gaussian functions:

$$v_1(x, y, 0) = 1.3 \, \exp\!\left(- \left(\frac{(x - 0.35)^2}{2 \cdot 0.12^2} + \frac{(y - 0.42)^2}{2 \cdot 0.10^2} \right) \right) + 0.02,$$

$$v_2(x, y, 0) = 1.4 \, \exp\!\left(- \left(\frac{(x - 0.68)^2}{2 \cdot 0.10^2} + \frac{(y - 0.60)^2}{2 \cdot 0.12^2} \right) \right) + 0.02.$$

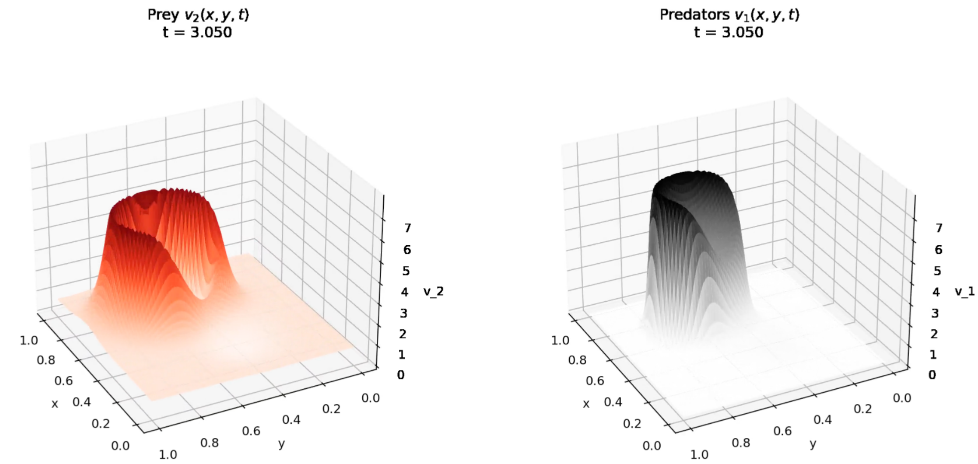

Figure 5: Solution profiles at $t = 3.050$ with homogeneous Neumann boundary conditions and localized Gaussian initial data.

For comparison, the next figure uses the same data with homogeneous Dirichlet conditions:

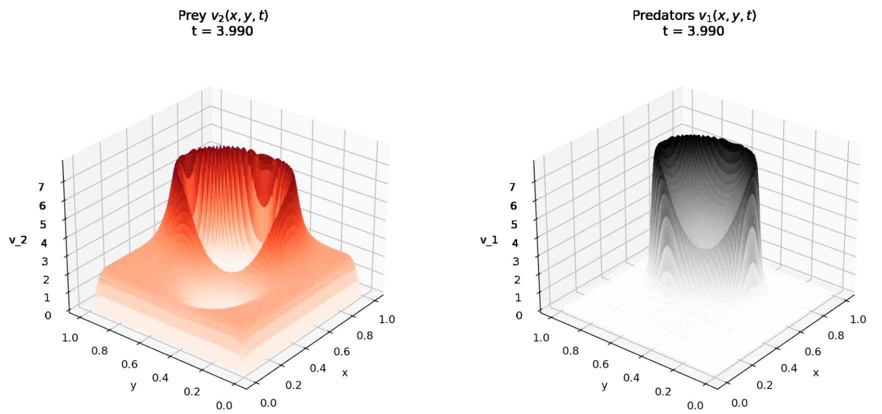

Figure 6: Solution profiles at $t = 3.990$ with homogeneous Dirichlet boundary conditions and localized Gaussian initial data.

2 Control Problems

We now pass from the simulation of uncontrolled dynamics to the design of a fishing policy. We first study the non-spatial system with a constant effort $u_2(t) = \bar{u}_2$, compute all equilibria, and determine their local stability. This calculation gives the threshold $q\bar{u}_2 = r$. We then allow u_2 to depend on (x, t), formulate a distributed optimal control problem, derive the state and adjoint equations, and approximate the optimality conditions numerically.

2.1 Model Without a Spatial Variable and Constant Controls

The controlled time-dependent model is the natural extension of the classical prey–predator system [14, 21]:

$$\begin{cases} \dot{v}_1(t) = v_1(t)\big(-m + \alpha v_2(t)\big) \\ \dot{v}_2(t) = v_2(t)\big(r - \beta v_1(t)\big) - q\, u_2(t)\, v_2(t) \end{cases}$$

with initial values $v_1(0) = v_1^0 \geq 0$ and $v_2(0) = v_2^0 \geq 0$. All parameters are positive. The coefficient m is the natural mortality rate of predators, α measures the increase of predator growth produced by prey, r is the prey growth rate without predators, β measures predation losses, and q converts the fishing effort u_2 into an additional prey mortality qu_2. Thus the controlled prey per-capita growth rate is $r - \beta v_1 - qu_2$. Assume $u_2(t) \geq 0$ and $(v_1^0, v_2^0) \in \mathbf{R}_+^2$. Since each right-hand side vanishes on its corresponding coordinate axis, the positive quadrant $\mathbf{R}_+^2$ is invariant. We first set $u_2(t) = \bar{u}_2 \geq 0$. An equilibrium (v_1^*, v_2^*) satisfies

$$\begin{cases} 0 = v_1^*(-m + \alpha v_2^*) \\ 0 = v_2^*(r - \beta v_1^*) - q\bar{u}_2\, v_2^* \end{cases}.$$

The extinction state $E_0 = (0,0)$ is always an equilibrium. If $v_1^* = 0$ and $v_2^* > 0$, the second equation gives $v_2^*(r - q\bar{u}_2) = 0$. Hence a positive predator-free equilibrium exists only in the critical case $r - q\bar{u}_2 = 0$. In that case, every point of the half-line $\{(0,s) : s \geq 0\}$ is stationary because prey growth and fishing mortality cancel exactly. If $r - q\bar{u}_2 \neq 0$, there is no positive predator-free equilibrium. For coexistence, both components are positive, so the two factors multiplying v_1^* and v_2^* must vanish:

$$-m + \alpha v_2^* = 0, \qquad r - \beta v_1^* - q\bar{u}_2 = 0,$$

which gives

$$v_2^* = \frac{m}{\alpha}, \qquad v_1^* = \frac{r - q\bar{u}_2}{\beta}.$$

Hence a positive coexistence equilibrium exists if and only if $r - q\bar{u}_2 > 0$, and then

$$E_* = \left(\frac{r - q\bar{u}_2}{\beta}, \frac{m}{\alpha}\right).$$

Remark. The coexistence condition is $q\bar{u}_2 < r$. When $q\bar{u}_2 \geq r$, fishing removes the positive coexistence equilibrium because prey growth no longer sustains the predator population.

Set $\rho := r - q\bar{u}_2$. The controlled system becomes

$$\begin{cases} \dot{v}_1 = v_1(-m + \alpha v_2) \\ \dot{v}_2 = v_2(\rho - \beta v_1) \end{cases}.$$

Assume first that $\rho = r - q\bar{u}_2 > 0$. The positive equilibrium is $E_* = (\rho/\beta, m/\alpha)$. The classical Lotka–Volterra Hamiltonian provides a direct stability argument [14]. Define

$$H(v_1, v_2) = \beta v_1 - \rho \ln v_1 + \alpha v_2 - m \ln v_2, \qquad \text{with } v_1 > 0,\ v_2 > 0.$$

For every trajectory in $(0, +\infty)^2$,

$$\frac{\mathrm{d}}{\mathrm{d}t} H(v_1(t), v_2(t)) = 0.$$

Indeed,

$$\frac{\partial H}{\partial v_1} = \beta - \frac{\rho}{v_1}, \qquad \frac{\partial H}{\partial v_2} = \alpha - \frac{m}{v_2}.$$

The chain rule gives

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} H(v_1(t), v_2(t)) &= \left(\beta - \frac{\rho}{v_1}\right)\dot{v}_1 + \left(\alpha - \frac{m}{v_2}\right)\dot{v}_2 \\ &= \left(\beta - \frac{\rho}{v_1}\right) v_1(-m + \alpha v_2) + \left(\alpha - \frac{m}{v_2}\right) v_2(\rho - \beta v_1) \\ &= (\beta v_1 - \rho)(-m + \alpha v_2) + (\alpha v_2 - m)(\rho - \beta v_1) \\ &= 0. \end{aligned}$$

Thus $H(v_1(t), v_2(t)) = H(v_1(0), v_2(0))$. Moreover, E_* is a strict local minimum of H, because

$$\frac{\partial H}{\partial v_1}(E_*) = 0, \qquad \frac{\partial H}{\partial v_2}(E_*) = 0,$$

and

$$\frac{\partial^2 H}{\partial v_1^2}(E_*) = \frac{\rho}{(v_1^*)^2} > 0, \qquad \frac{\partial^2 H}{\partial v_2^2}(E_*) = \frac{m}{(v_2^*)^2} > 0, \qquad \frac{\partial^2 H}{\partial v_1 \partial v_2} = 0.$$

The nearby level sets $\{H = c\}$ are closed curves around E_*. A trajectory stays on one such curve, so nearby solutions remain nearby and oscillate around the equilibrium. Therefore E_* is a center: it is Lyapunov stable and not asymptotically stable.

At the extinction state, the Jacobian is

$$J(E_0) = \begin{pmatrix} -m & 0 \\ 0 & r - q\bar{u}_2 \end{pmatrix},$$

Its eigenvalues are $-m$ and $r - q\bar{u}_2$. Together with the equilibrium calculation, this gives the three regimes

- $r - q\bar{u}_2 < 0$: $E_0 = (0, 0)$ is locally asymptotically stable,
- $r - q\bar{u}_2 = 0$: the equilibrium set is $\{(0, s) : s \geq 0\}$,
- $r - q\bar{u}_2 > 0$: $E_* = ((r - q\bar{u}_2)/\beta, m/\alpha)$ exists and is a center, so it is locally stable and not asymptotically stable.

In the critical case $r - q\bar{u}_2 = 0$, natural prey growth and fishing mortality balance exactly when $v_1 = 0$. Indeed, the prey equation reduces to

$$\dot{v}_2 = (r - q\bar{u}_2)v_2.$$

Thus $r - q\bar{u}_2 = 0$ gives $\dot{v}_2 = 0$ in the absence of predators. If $v_1 > 0$, the prey equation becomes

$$\dot{v}_2 = -\beta v_1 v_2,$$

and therefore $\dot{v}_2 < 0$ whenever $v_1 v_2 > 0$.

The Python experiment [2] illustrates the three regimes with the same initial condition. We select three constant efforts $\bar{u}_2$: one with $r - q\bar{u}_2 < 0$, one with $r - q\bar{u}_2 = 0$, and one with $r - q\bar{u}_2 > 0$. The resulting trajectories show extinction, the critical balance, and closed coexistence oscillations.

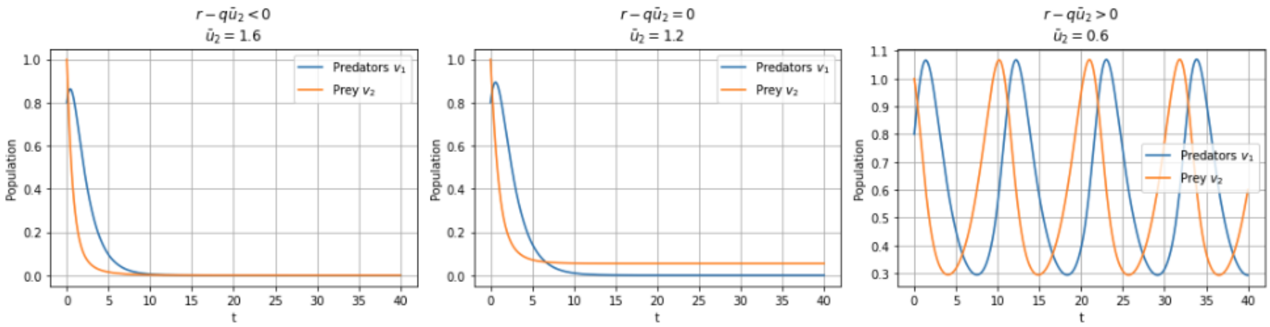

Figure 7: The three regimes of the controlled Lotka–Volterra ODE: extinction for $r - q\bar{u}_2 < 0$, critical predator-free balance for $r - q\bar{u}_2 = 0$, and coexistence oscillations for $r - q\bar{u}_2 > 0$.

2.2 Space-time Optimal Control Problem

We now let the fishing effort depend on both variables, $u_2 = u_2(x, t)$. The formulation follows optimal control for evolution equations [12], PDE-constrained optimization [20], and Pontryagin's principle [19]. The harvested prey quantity is $\int_Q q\, u_2 v_2 \, dx \, dt$, so maximizing yield is equivalent to minimizing its negative. The state is defined on $Q = \Omega \times (0, T)$, where $\Omega \subseteq \mathbf{R}^d$ is bounded. Both populations diffuse, with $\mu_1, \mu_2 > 0$, and fishing adds the mortality term $-qu_2v_2$ to the prey equation. The controlled system is

$$\begin{cases} \partial_t v_1(x,t) - \mu_1 \Delta v_1(x,t) = v_1(x,t)\big(-m + \alpha v_2(x,t)\big), & \text{in } \Omega \times (0,T), \\ \partial_t v_2(x,t) - \mu_2 \Delta v_2(x,t) = v_2(x,t)\big(r - \beta v_1(x,t)\big) - q\, u_2(x,t)\, v_2(x,t), & \text{in } \Omega \times (0,T), \\ \dfrac{\partial v_1}{\partial n} = \dfrac{\partial v_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0,T), \\ v_1(\cdot, 0) = v_1^0, \qquad v_2(\cdot, 0) = v_2^0, & \text{in } \Omega. \end{cases}$$

The value $u_2(x, t)$ measures the local fishing effort exerted on prey at $(x, t) \in Q$. We prescribe target densities $v_1^{\text{tar}}, v_2^{\text{tar}} \in L^2(Q)$. They may vary in space and time; a stationary target is simply independent of t. Physical and operational limits are represented by a pointwise upper bound $U_2^{\max} > 0$. The admissible control set is

$$\mathscr{U}_{\text{ad}}^{\Omega} := \{u_2 \in L^\infty(Q) \mid 0 \leq u_2(x,t) \leq U_2^{\max} \text{ a.e. in } Q\}.$$

For $(v_1, v_2, u_2) \in L^2(Q)^3$, define

$$\mathscr{J} : L^2(Q) \times L^2(Q) \times L^2(Q) \longrightarrow \mathbf{R}$$

by

$$\mathscr{J}(v_1, v_2, u_2) := \frac{1}{2}\int_0^T \int_\Omega \Big[\gamma_1\big(v_1(x,t) - v_1^{\mathrm{tar}}(x,t)\big)^2 + \gamma_2\big(v_2(x,t) - v_2^{\mathrm{tar}}(x,t)\big)^2 + \lambda_2 u_2(x,t)^2\Big]\,\mathrm{d}x\,\mathrm{d}t$$
$$- \eta \int_0^T \int_\Omega q\, u_2(x,t)\, v_2(x,t)\,\mathrm{d}x\,\mathrm{d}t.$$

Let $S(u_2) = (v_1, v_2)$ be the state generated by u_2. The reduced functional is

$$J : \mathscr{U}_{\mathrm{ad}}^{\Omega} \longrightarrow \mathbf{R}, \qquad J(u_2) := \mathscr{J}(S(u_2), u_2).$$

The three terms have the following roles:

- $\gamma_i > 0$ weights the tracking error $\|v_i - v_i^{\mathrm{tar}}\|^2_{L^2(Q)}$, $i = 1, 2$,
- $\lambda_2 > 0$ weights the effort $\|u_2\|^2_{L^2(Q)}$,
- $-\eta q \int_Q u_2 v_2 \,\mathrm{d}x\,\mathrm{d}t$ rewards the harvested quantity.

Remark. Minimizing $J = J_{\text{track}} + J_{\text{effort}} - J_{\text{yield}}$ balances population targets, fishing cost, and yield.

Before minimizing J, we verify that each admissible control produces one state on the whole interval $[0, T]$. We work with weak solutions in the standard parabolic setting [6]; the semilinear theory and the qualitative properties used below are also developed in [9, 18]. A bounded Lipschitz domain is sufficient. The initial values are understood in the usual trace sense, and the homogeneous Neumann condition is included in the variational formulation, so no separate boundary trace of ∇v_i is required. For a fixed $u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega}$, we seek states with the regularity

$$v_i \in L^2(0, T; H^1(\Omega)) \cap L^\infty(Q), \qquad \partial_t v_i \in L^2(0, T; H^{-1}(\Omega)), \qquad i = 1, 2.$$

The equations hold in $L^2(0, T; H^{-1}(\Omega))$. For $i \in \{1, 2\}$, the diffusion term and the weak Neumann condition are encoded by

$$\langle \partial_t v_i, \varphi\rangle_{H^{-1}, H^1} + \mu_i \int_\Omega \nabla v_i \cdot \nabla\varphi \,\mathrm{d}x = \int_\Omega F_i\, \varphi \,\mathrm{d}x, \qquad \forall \varphi \in H^1(\Omega),$$

for every $\varphi \in H^1(\Omega)$ and for a.e. $t \in (0, T)$, where F_i is the reaction term of species i.

Proposition 2.1 (Well-posed state system)

Let $\Omega \subseteq \mathbf{R}^d$ be bounded and Lipschitz, set $Q = \Omega \times (0, T)$, and assume $\mu_1, \mu_2, m, \alpha, r, \beta, q > 0$. Let

$$u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega} = \{u_2 \in L^\infty(Q) \mid 0 \le u_2(x,t) \le U_2^{\max} \text{ a.e. in } Q\}.$$

Assume $v_1^0, v_2^0 \in L^\infty(\Omega)$ and $v_1^0, v_2^0 \ge 0$ a.e. in Ω. Then the system

$$\begin{cases} \partial_t v_1 - \mu_1 \Delta v_1 = v_1(-m + \alpha v_2), & \text{in } Q,\\ \partial_t v_2 - \mu_2 \Delta v_2 = v_2(r - \beta v_1) - q u_2 v_2, & \text{in } Q,\\ \dfrac{\partial v_1}{\partial n} = \dfrac{\partial v_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0, T),\\ v_1(\cdot, 0) = v_1^0, \qquad v_2(\cdot, 0) = v_2^0, & \text{in } \Omega, \end{cases}$$

has a unique bounded weak solution on every finite interval $[0, T]$, with

$$v_i \in L^2(0, T; H^1(\Omega)) \cap L^\infty(Q), \qquad \partial_t v_i \in L^2(0, T; H^{-1}(\Omega)), \qquad i = 1, 2.$$

Moreover, $(v_1, v_2) \in L^\infty(Q; \mathbf{R}^2_+)$; that is, $v_1, v_2 \ge 0$ a.e. in Q.

Proof. We give the main steps. Standard semilinear parabolic theory provides local existence, uniqueness on bounded sets, and a continuation criterion; see [1, 9]. The weak comparison principle used for nonnegativity is recalled in [18]. Fix $u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega}$. To apply these results, write the system in the form

$$\partial_t v_1 - \mu_1 \Delta v_1 = f_1(v_1, v_2), \qquad \partial_t v_2 - \mu_2 \Delta v_2 = f_2(x, t, v_1, v_2),$$

where

$$\begin{cases} f_1(v_1, v_2) := v_1(-m + \alpha v_2)\\ f_2(x, t, v_1, v_2) := v_2(r - \beta v_1) - q u_2(x,t) v_2. \end{cases}$$

Because $u_2 \in L^\infty(Q)$, f_1 and f_2 are locally Lipschitz in (v_1, v_2) on every bounded subset of $\mathbb{R}^2$, uniformly in (x, t). The local existence theorem for semilinear parabolic systems with Neumann conditions [1] gives a maximal weak solution on $[0, T_{\max})$ such that

$$v_i \in L^2(0, \tau; H^1(\Omega)) \cap L^\infty(\Omega \times (0, \tau)), \qquad \partial_t v_i \in L^2(0, \tau; H^{-1}(\Omega)),$$

for every $\tau < T_{\max}$ and $i = 1, 2$. We next prove positivity. Since $f_1(0, v_2) = 0$ and $f_2(x, t, v_1, 0) = 0$, both coordinate axes are invariant. Equivalently, test the equations with the negative parts

$$v_1^- := \max(-v_1, 0), \qquad v_2^- := \max(-v_2, 0),$$

The Neumann boundary terms vanish, and the resulting inequalities control v_1^- and v_2^-. Since $v_1^0, v_2^0 \geq 0$, the weak maximum principle [6, 18] gives $(v_1, v_2) \geq (0, 0)$ a.e. on $\Omega \times (0, T_{\max})$. We now obtain L^∞ bounds. From $v_1, u_2 \geq 0$,

$$r - \beta v_1 - q u_2 \leq r.$$

Thus

$$\partial_t v_2 - \mu_2 \Delta v_2 = v_2(r - \beta v_1 - q u_2) \leq r v_2.$$

Compare v_2 with $z' = rz$, $z(0) = \|v_2^0\|_{L^\infty(\Omega)}$. For $0 < t < T_{\max}$ this gives

$$0 \leq v_2(x, t) \leq \|v_2^0\|_{L^\infty(\Omega)} e^{rt} \quad \text{for a.e. } x \in \Omega.$$

Hence, for each $\tau < T_{\max}$,

$$\|v_2\|_{L^\infty(\Omega \times (0,\tau))} \leq \|v_2^0\|_{L^\infty(\Omega)} e^{r\tau}.$$

Define $M_2(\tau) := \|v_2^0\|_{L^\infty(\Omega)} e^{r\tau}$. Then

$$0 \leq v_2(x, t) \leq M_2(\tau) \quad \text{for a.e. } (x, t) \in \Omega \times (0, \tau).$$

The first state equation now gives

$$\partial_t v_1 - \mu_1 \Delta v_1 = v_1(-m + \alpha v_2) \leq \alpha M_2(\tau) v_1 \quad \text{in } \Omega \times (0, \tau).$$

Compare this inequality with $y' = \alpha M_2(\tau) y$, $y(0) = \|v_1^0\|_{L^\infty(\Omega)}$. We obtain

$$0 \leq v_1(x, t) \leq \|v_1^0\|_{L^\infty(\Omega)} e^{\alpha M_2(\tau) t} \quad \text{for a.e. } (x, t) \in \Omega \times (0, \tau).$$

Therefore

$$\|v_1\|_{L^\infty(\Omega \times (0,\tau))} \leq \|v_1^0\|_{L^\infty(\Omega)} e^{\alpha M_2(\tau)\tau}.$$

These estimates prevent finite-time blow-up in L^∞. Indeed, if $T_{\max} < +\infty$, the continuation criterion for semilinear parabolic systems would require

$$\limsup_{t \to T_{\max}^-} \left(\|v_1(t)\|_{L^\infty(\Omega)} + \|v_2(t)\|_{L^\infty(\Omega)} \right) = +\infty.$$

The estimates above exclude this blow-up alternative, so $T_{\max} = +\infty$ and the solution exists on the prescribed interval $[0, T]$. It remains to prove uniqueness. Let (v_1, v_2) and (w_1, w_2) be two bounded weak solutions associated with the same control and the same initial data, and set $z_1 = v_1 - w_1$, $z_2 = v_2 - w_2$. Both solutions remain in a common bounded subset of $\mathbb{R}^2$ for a.e. $(x, t) \in Q$. On this set, the nonlinearities f_1 and f_2 have a common Lipschitz constant. Consequently, there is $L_T > 0$ such that, for a.e. $t \in (0, T)$,

$$\|f_1(v_1, v_2) - f_1(w_1, w_2)\|_{L^2(\Omega)} + \|f_2(\cdot, t, v_1, v_2) - f_2(\cdot, t, w_1, w_2)\|_{L^2(\Omega)} \leq L_T \left(\|z_1(t)\|_{L^2(\Omega)} + \|z_2(t)\|_{L^2(\Omega)} \right).$$

Subtract the two systems, test the equations with z_1 and z_2, integrate over Ω, and use the weak Neumann conditions. This gives

$$\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \left(\|z_1(t)\|_{L^2(\Omega)}^2 + \|z_2(t)\|_{L^2(\Omega)}^2 \right) + \mu_1 \|\nabla z_1(t)\|_{L^2(\Omega)}^2 + \mu_2 \|\nabla z_2(t)\|_{L^2(\Omega)}^2 \leq C_T \left(\|z_1(t)\|_{L^2(\Omega)}^2 + \|z_2(t)\|_{L^2(\Omega)}^2 \right),$$

for a constant $C_T > 0$. Since the diffusion terms are nonnegative, we may omit them and obtain

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(\|z_1(t)\|_{L^2(\Omega)}^2 + \|z_2(t)\|_{L^2(\Omega)}^2 \right) \leq C_T \left(\|z_1(t)\|_{L^2(\Omega)}^2 + \|z_2(t)\|_{L^2(\Omega)}^2 \right),$$

The value of C_T may change from one estimate to the next. Since $z_1(0) = z_2(0) = 0$, Gronwall's lemma gives $z_1 = z_2 = 0$ in Q. The weak solution is therefore unique. □

Proposition 2.2 (Directional differentiability under a local stability assumption)

Fix $u_2 \in \mathscr{U}_{\rm ad}^{\Omega}$ and let $(v_1, v_2) = S(u_2) \in L^\infty(Q)^2$. We assume a local Lipschitz stability estimate for the control-to-state map. Thus, for some neighbourhood $\mathscr{V}$ of u_2 in $L^\infty(Q)$ and some $C > 0$, every $\widetilde{u}_2 \in \mathscr{V} \cap \mathscr{U}_{\rm ad}^{\Omega}$ has a state close to $S(u_2)$ and satisfies

$$\|S(\widetilde{u}_2) - S(u_2)\|_{L^\infty(Q)^2} \le C\|\widetilde{u}_2 - u_2\|_{L^\infty(Q)}.$$

This estimate is standard under the boundedness assumptions used here; see [20].
Let $h \in L^\infty(Q)$ be admissible, so there is $\varepsilon_0 > 0$ such that

$$u_2 + \varepsilon h \in \mathscr{U}_{\rm ad}^{\Omega} \quad \text{for all } 0 \le \varepsilon < \varepsilon_0.$$

Then the control-to-state map

$$S : \mathscr{U}_{\rm ad}^{\Omega} \longrightarrow L^2(Q)^2, \qquad u_2 \longmapsto (v_1, v_2),$$

is directionally differentiable at u_2 along h. The derivative $S'(u_2)h = (z_1, z_2)$ is the unique weak solution of

$$\begin{cases} \partial_t z_1 - \mu_1 \Delta z_1 = (-m + \alpha v_2) z_1 + \alpha v_1 z_2, & \text{in } Q, \\ \partial_t z_2 - \mu_2 \Delta z_2 = -\beta v_2 z_1 + (r - \beta v_1 - q u_2) z_2 - q v_2 h, & \text{in } Q, \\ \dfrac{\partial z_1}{\partial n} = \dfrac{\partial z_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0, T), \\ z_1(\cdot, 0) = 0, \qquad z_2(\cdot, 0) = 0, & \text{in } \Omega. \end{cases} \tag{5}$$

Equivalently,

$$\frac{S(u_2 + \varepsilon h) - S(u_2)}{\varepsilon} \underset{\varepsilon \to 0^+}{\longrightarrow} S'(u_2)h \quad \text{in } L^2(Q)^2.$$

Proof. For $0 < \varepsilon < \varepsilon_0$ small, local stability gives

$$\|S(u_2 + \varepsilon h) - S(u_2)\|_{L^\infty(Q)^2} \le C|\varepsilon|\|h\|_{L^\infty(Q)}.$$

Set $u_2^\varepsilon = u_2 + \varepsilon h$ and $(v_1^\varepsilon, v_2^\varepsilon) = S(u_2^\varepsilon)$. Define $(v_1^\varepsilon, v_2^\varepsilon) = S(u_2^\varepsilon)$ be the corresponding state. We define the difference quotients

$$Z_i^\varepsilon := \frac{v_i^\varepsilon - v_i}{\varepsilon}, \qquad i = 1, 2.$$

Subtract the state system for (v_1, v_2) from the one for $(v_1^\varepsilon, v_2^\varepsilon)$ and divide by ε. Then

$$\begin{cases} \partial_t Z_1^\varepsilon - \mu_1 \Delta Z_1^\varepsilon = \dfrac{v_1^\varepsilon(-m + \alpha v_2^\varepsilon) - v_1(-m + \alpha v_2)}{\varepsilon}, \\ \partial_t Z_2^\varepsilon - \mu_2 \Delta Z_2^\varepsilon = \dfrac{v_2^\varepsilon(r - \beta v_1^\varepsilon) - q u_2^\varepsilon v_2^\varepsilon - v_2(r - \beta v_1) + q u_2 v_2}{\varepsilon}. \end{cases}$$

Using

$$v_i^\varepsilon = v_i + \varepsilon Z_i^\varepsilon, \qquad u_2^\varepsilon = u_2 + \varepsilon h,$$

The first quotient becomes

$$(-m + \alpha v_2) Z_1^\varepsilon + \alpha v_1 Z_2^\varepsilon + \varepsilon \alpha Z_1^\varepsilon Z_2^\varepsilon.$$

The second quotient is

$$-\beta v_2 Z_1^\varepsilon + (r - \beta v_1 - q u_2) Z_2^\varepsilon - q v_2 h - \varepsilon \beta Z_1^\varepsilon Z_2^\varepsilon - \varepsilon q h Z_2^\varepsilon.$$

Thus $(Z_1^\varepsilon, Z_2^\varepsilon)$ satisfies the linearized system (5), with additional quadratic remainders multiplied by ε. We first control the difference quotients uniformly. For small $\varepsilon > 0$, the states v_i^ε and v_i remain in a common bounded subset of $L^\infty(Q)$. The reaction maps are Lipschitz on this set. Applying the standard stability estimate for semilinear parabolic systems [6, 9, 20] to $v_i^\varepsilon - v_i$ gives

$$\|v_1^\varepsilon - v_1\|_{L^\infty(Q)} + \|v_2^\varepsilon - v_2\|_{L^\infty(Q)} \le C|\varepsilon|\|h\|_{L^\infty(Q)}.$$

Thus,

$$\|Z_1^\varepsilon\|_{L^\infty(Q)} + \|Z_2^\varepsilon\|_{L^\infty(Q)} \le C\|h\|_{L^\infty(Q)},$$

The constant C is independent of ε for $0 < \varepsilon < \varepsilon_0$ small. Thus $(Z_1^\varepsilon, Z_2^\varepsilon)$ is uniformly bounded in $L^\infty(Q)^2$, and hence also in $L^2(Q)^2$. The coefficients of the linearized system (5) belong to $L^\infty(Q)$, so standard linear parabolic theory gives a unique weak solution (z_1, z_2). We now compare this solution with the difference quotients. Set $R_i^\varepsilon = Z_i^\varepsilon - z_i$,

$i = 1, 2$. The functions R_i^ε have zero initial data and homogeneous Neumann conditions. Subtracting (5) from the equations satisfied by Z_i^ε gives

$$\partial_t R_1^\varepsilon - \mu_1 \Delta R_1^\varepsilon = (-m + \alpha v_2) R_1^\varepsilon + \alpha v_1 R_2^\varepsilon + \varepsilon \alpha Z_1^\varepsilon Z_2^\varepsilon,$$

and

$$\partial_t R_2^\varepsilon - \mu_2 \Delta R_2^\varepsilon = -\beta v_2 R_1^\varepsilon + (r - \beta v_1 - q u_2) R_2^\varepsilon - \varepsilon \beta Z_1^\varepsilon Z_2^\varepsilon - \varepsilon q h Z_2^\varepsilon,$$

with homogeneous Neumann conditions and

$$R_1^\varepsilon(\cdot, 0) = R_2^\varepsilon(\cdot, 0) = 0.$$

The uniform L^∞ bounds for Z_1^ε and Z_2^ε imply

$$\|\varepsilon Z_1^\varepsilon Z_2^\varepsilon\|_{L^2(Q)} \leq C|\varepsilon| \|h\|_{L^\infty(Q)}^2, \quad \|\varepsilon h Z_2^\varepsilon\|_{L^2(Q)} \leq C|\varepsilon| \|h\|_{L^\infty(Q)}^2.$$

Test the remainder equations with R_1^ε and R_2^ε, integrate over Ω, and use the Neumann conditions. We obtain R_1^ε and R_2^ε, respectively, integrating over Ω, and using the homogeneous Neumann boundary conditions, we obtain

$$\begin{aligned} \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left(\|R_1^\varepsilon(t)\|_{L^2(\Omega)}^2 + \|R_2^\varepsilon(t)\|_{L^2(\Omega)}^2\right) + \mu_1 \|\nabla R_1^\varepsilon(t)\|_{L^2(\Omega)}^2 + \mu_2 \|\nabla R_2^\varepsilon(t)\|_{L^2(\Omega)}^2 \\ \leq C\left(\|R_1^\varepsilon(t)\|_{L^2(\Omega)}^2 + \|R_2^\varepsilon(t)\|_{L^2(\Omega)}^2\right) + C\varepsilon^2. \end{aligned}$$

Here $C > 0$ depends on the state bounds, the model parameters, and $\|h\|_{L^\infty(Q)}$, but it is independent of ε. After removing the nonnegative diffusion terms and using the zero initial data, Gronwall's lemma gives

$$\|R_1^\varepsilon\|_{L^\infty(0,T;L^2(\Omega))}^2 + \|R_2^\varepsilon\|_{L^\infty(0,T;L^2(\Omega))}^2 \leq C\varepsilon^2.$$

Therefore $R_i^\varepsilon \to 0$ in $L^2(Q)$ and $Z_i^\varepsilon \to z_i$ in $L^2(Q)$ for $i = 1, 2$. This proves the directional differentiability of S along h and identifies $S'(u_2)h$. □

We now seek $u_2^* \in \mathscr{U}_{\mathrm{ad}}^\Omega$ such that

$$J(u_2^*) = \inf_{u_2 \in \mathscr{U}_{\mathrm{ad}}^\Omega} J(u_2),$$

under the constraint

$$\begin{cases} \partial_t v_1 - \mu_1 \Delta v_1 = v_1(-m + \alpha v_2) \\ \partial_t v_2 - \mu_2 \Delta v_2 = v_2(r - \beta v_1) - q u_2 v_2 \end{cases}$$

with nonnegative initial data $v_1(\cdot, 0) = v_1^0$ and $v_2(\cdot, 0) = v_2^0$ in Ω, together with $\partial_n v_1 = \partial_n v_2 = 0$ on $\partial\Omega \times (0, T)$. We first prove that a minimizer exists.

Proposition 2.3 (Existence of an optimal control)

Let $\Omega \subseteq \mathbb{R}^d$ be bounded with sufficiently smooth boundary and set $Q = \Omega \times (0, T)$. Assume that all diffusion and reaction parameters, as well as $\gamma_1, \gamma_2, \lambda_2$, are positive, and let $\eta \geq 0$. Suppose $v_1^0, v_2^0 \in L^\infty(\Omega)$ are nonnegative a.e. and $v_1^{\mathrm{tar}}, v_2^{\mathrm{tar}} \in L^2(Q)$.

Then there exists an admissible control

$$u_2^* \in \mathscr{U}_{\mathrm{ad}}^\Omega = \{u_2 \in L^\infty(Q) \mid 0 \leq u_2(x, t) \leq U_2^{\max} \text{ a.e. in } Q\}$$

such that

$$J(u_2^*) = \inf_{u_2 \in \mathscr{U}_{\mathrm{ad}}^\Omega} J(u_2).$$

Proof. We use the direct method of the calculus of variations [12, 20]. Proposition 2.1 assigns to each $u_2 \in \mathscr{U}_{\mathrm{ad}}^\Omega$ a unique nonnegative state $(v_1, v_2) \in L^\infty(Q)^2$. Since Q has finite measure and the targets belong to $L^2(Q)$, every tracking, effort, and yield term in

$$J(u_2) = \frac{1}{2}\int_Q \left[\gamma_1(v_1 - v_1^{\mathrm{tar}})^2 + \gamma_2(v_2 - v_2^{\mathrm{tar}})^2 + \lambda_2 u_2^2\right] \mathrm{d}x\, \mathrm{d}t - \eta \int_Q q u_2 v_2 \, \mathrm{d}x\, \mathrm{d}t$$

are finite. We also need a lower bound. The tracking and effort terms are nonnegative, and only the yield term can be negative. Since $0 \leq u_2 \leq U_2^{\max}$ and

$$0 \leq v_2(x, t) \leq M_2(T) := \|v_2^0\|_{L^\infty(\Omega)} e^{rT} \quad \text{a.e. in } Q,$$

we obtain

$$-\eta \int_Q q u_2 v_2 \,\mathrm{d}x\,\mathrm{d}t \geq -\eta q U_2^{\max} M_2(T)|Q|.$$

Consequently,

$$J(u_2) \geq -\eta q U_2^{\max} M_2(T)|Q|$$

for every $u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega}$. Choose a minimizing sequence $(u_2^k)_{k\geq 1} \subset \mathscr{U}_{\mathrm{ad}}^{\Omega}$, so

$$J(u_2^k) \longrightarrow \inf_{u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega}} J(u_2).$$

The box constraint gives $\|u_2^k\|_{L^\infty(Q)} \leq U_2^{\max}$ for every k. By the Banach–Alaoglu theorem [6], the closed ball of $L^\infty(Q)$ is weakly-* compact. After extracting a subsequence, there is $u_2^* \in L^\infty(Q)$ such that

$$u_2^k \overset{*}{\rightharpoonup} u_2^* \quad \text{weakly-* in } L^\infty(Q).$$

The admissible set is weakly-* closed in $L^\infty(Q)$, so the limit u_2^* still satisfies $0 \leq u_2^* \leq U_2^{\max}$. Let $(v_1^k, v_2^k) = S(u_2^k)$. The comparison estimates from Proposition 2.1 are uniform in k because the same control bound holds for the whole sequence. Standard energy estimates for the two parabolic equations then give, for $i = 1, 2$,

$$(v_i^k)_{k\geq 1} \quad \text{bounded in } L^2(0,T;H^1(\Omega)), \quad \text{and} \quad (\partial_t v_i^k)_{k\geq 1} \quad \text{bounded in } L^2(0,T;H^{-1}(\Omega)).$$

The reaction terms are uniformly bounded in $L^2(Q)$ because the controls and the two state components are uniformly bounded in $L^\infty(Q)$. The Aubin–Lions lemma [6, 20] therefore gives, after extraction of a subsequence, functions $\bar{v}_1, \bar{v}_2 \in L^2(Q)$ such that $v_i^k \to \bar{v}_i$ strongly in $L^2(Q)$ for $i = 1, 2$. The same uniform estimates give $v_i^k \overset{*}{\rightharpoonup} \bar{v}_i$ weakly-* in $L^\infty(Q)$. Since every v_i^k is nonnegative, $\bar{v}_i \geq 0$ a.e. Strong L^2 convergence and the L^∞ bound also imply convergence in $L^p(Q)$ for every finite $p \geq 2$ along this subsequence. In particular, $v_1^k v_2^k \to \bar{v}_1\bar{v}_2$ strongly in $L^1(Q)$, which is enough to pass to the bilinear reaction terms. For the control term,

$$u_2^k v_2^k \rightharpoonup u_2^* \bar{v}_2 \quad \text{in the sense of distributions.}$$

Indeed,

$$u_2^k v_2^k - u_2^* \bar{v}_2 = u_2^k (v_2^k - \bar{v}_2) + (u_2^k - u_2^*)\bar{v}_2.$$

The first term tends to zero in $L^1(Q)$ because (u_2^k) is bounded in $L^\infty(Q)$ and $v_2^k \to \bar{v}_2$ in $L^1(Q)$. For the second term, take $\varphi \in C_c^\infty(Q)$. Since $\bar{v}_2\varphi \in L^1(Q)$,

$$\int_Q (u_2^k - u_2^*)\bar{v}_2 \varphi \,\mathrm{d}x\,\mathrm{d}t \longrightarrow 0$$

by weak-* convergence in $L^\infty(Q)$. Hence $(\bar{v}_1, \bar{v}_2)$ solves the state system for u_2^*:

$$\begin{cases} \partial_t \bar{v}_1 - \mu_1 \Delta \bar{v}_1 = \bar{v}_1(-m + \alpha \bar{v}_2), & \text{in } Q, \\ \partial_t \bar{v}_2 - \mu_2 \Delta \bar{v}_2 = \bar{v}_2(r - \beta \bar{v}_1) - q u_2^* \bar{v}_2, & \text{in } Q, \\ \dfrac{\partial \bar{v}_1}{\partial n} = \dfrac{\partial \bar{v}_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0,T), \\ \bar{v}_1(\cdot,0) = v_1^0, \qquad \bar{v}_2(\cdot,0) = v_2^0, & \text{in } \Omega. \end{cases}$$

By uniqueness of the state equation, the limit pair is precisely $S(u_2^*) = (\bar{v}_1, \bar{v}_2)$; no second state can be associated with the same limit control. We now pass to the four terms of the cost. Since $v_i^k \to \bar{v}_i$ strongly in $L^2(Q)$, $i = 1, 2$,

$$\int_Q (v_i^k - v_i^{\mathrm{tar}})^2 \,\mathrm{d}x\,\mathrm{d}t \longrightarrow \int_Q (\bar{v}_i - v_i^{\mathrm{tar}})^2 \,\mathrm{d}x\,\mathrm{d}t, \qquad i = 1, 2.$$

The sequence (u_2^k) is bounded in $L^\infty(Q)$ and therefore in $L^2(Q)$. Its weak-* limit in $L^\infty(Q)$ is also its weak limit in $L^2(Q)$. By weak lower semicontinuity of the squared L^2 norm,

$$\int_Q (u_2^*)^2 \,\mathrm{d}x\,\mathrm{d}t \leq \liminf_{k\to+\infty} \int_Q (u_2^k)^2 \,\mathrm{d}x\,\mathrm{d}t.$$

It remains to pass to the limit in the yield. Write

$$u_2^k v_2^k - u_2^* \bar{v}_2 = u_2^k (v_2^k - \bar{v}_2) + (u_2^k - u_2^*)\bar{v}_2.$$

The first term converges to zero in $L^1(Q)$ because $0 \le u_2^k \le U_2^{\max}$ and $v_2^k \to \bar{v}_2$ strongly in $L^1(Q)$. For the second term, the weak-* convergence of u_2^k can be tested against every function in $L^1(Q)$. Since $\bar{v}_2 \in L^1(Q)$, it follows that

$$\int_Q (u_2^k - u_2^*)\bar{v}_2 \,\mathrm{d}x\,\mathrm{d}t \longrightarrow 0.$$

Therefore

$$\int_Q q u_2^k v_2^k \,\mathrm{d}x\,\mathrm{d}t \longrightarrow \int_Q q u_2^* \bar{v}_2 \,\mathrm{d}x\,\mathrm{d}t.$$

Combining the three parts of the cost,

$$J(u_2^*) \le \liminf_{k\to+\infty} J(u_2^k).$$

Since (u_2^k) is minimizing,

$$\lim_{k\to+\infty} J(u_2^k) = \inf_{u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega}} J(u_2).$$

Hence

$$J(u_2^*) \le \inf_{u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega}} J(u_2).$$

Since $u_2^* \in \mathscr{U}_{\mathrm{ad}}^{\Omega}$, the reverse inequality also holds:

$$J(u_2^*) \ge \inf_{u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega}} J(u_2).$$

We conclude that

$$J(u_2^*) = \inf_{u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega}} J(u_2),$$

so u_2^* is an optimal control. □

Remark. The compactness argument proves that at least one optimal control exists [12, 20]. The control–state map is nonlinear, so the reduced problem is generally nonconvex and the minimizer need not be unique. The optimality system below gives necessary first-order conditions. A forward–backward sweep computes a stationary solution of this system; it does not by itself prove global minimality.

We now derive the first-order optimality system. It is important to keep the linearized state variables (z_1, z_2) separate from the adjoint variables (p_1, p_2), because they solve equations in opposite time directions and have different data. Set $y = (v_1, v_2)$. We place diffusion in the linear operator A and collect the local reactions and fishing term in F. The state system then has the compact form

$$\partial_t y = Ay + F(x, t, y, u_2), \qquad y = (v_1, v_2),$$

with

$$Ay = \begin{pmatrix} \mu_1 \Delta v_1 \\ \mu_2 \Delta v_2 \end{pmatrix}, \qquad F(x, t, v_1, v_2, u_2) = \begin{pmatrix} v_1(-m + \alpha v_2) \\ v_2(r - \beta v_1) - q u_2 v_2 \end{pmatrix}.$$

Let L be the running cost and $\mathscr{L}$ the augmented Lagrangian. The integrand of J is

$$L(x, t, v_1, v_2, u_2) = \frac{1}{2}\left[\gamma_1 (v_1 - v_1^{\mathrm{tar}})^2 + \gamma_2 (v_2 - v_2^{\mathrm{tar}})^2 + \lambda_2 u_2^2\right] - \eta q u_2 v_2.$$

For the adjoint $p = (p_1, p_2)$ [12, 19, 20], define the local Hamiltonian density by

$$\mathscr{H}(x, t, v_1, v_2, u_2, p_1, p_2) = L(x, t, v_1, v_2, u_2) + p \cdot F(x, t, v_1, v_2, u_2),$$

or, in full,

$$\begin{aligned} \mathscr{H}(x, t, v_1, v_2, u_2, p_1, p_2) := \frac{1}{2}&\left[\gamma_1 (v_1 - v_1^{\mathrm{tar}})^2 + \gamma_2 (v_2 - v_2^{\mathrm{tar}})^2 + \lambda_2 u_2^2\right] - \eta q u_2 v_2 \\ &+ p_1 v_1(-m + \alpha v_2) + p_2\big(v_2(r - \beta v_1) - q u_2 v_2\big). \end{aligned}$$

The local Hamiltonian contains the running cost, reactions, and control terms. We keep diffusion outside $\mathscr{H}$. The augmented Lagrangian is

$$\begin{aligned} \mathscr{L}(v_1, v_2, u_2, p_1, p_2) := \mathscr{J}(v_1, v_2, u_2) &- \int_Q p_1 \Big(\partial_t v_1 - \mu_1 \Delta v_1 - v_1(-m + \alpha v_2)\Big) \,\mathrm{d}x\,\mathrm{d}t \\ &- \int_Q p_2 \Big(\partial_t v_2 - \mu_2 \Delta v_2 - v_2(r - \beta v_1) + q u_2 v_2\Big) \,\mathrm{d}x\,\mathrm{d}t. \end{aligned}$$

To derive the adjoint equations from $\mathscr{L}$ [12, 20], take sufficiently regular variations δv_1 and δv_2. Since the initial state is fixed, $\delta v_i(\cdot,0)=0$. Since the Neumann boundary condition is also fixed, the variations satisfy $\partial_n \delta v_i = 0$. We compute the Gâteaux derivative of $\mathscr{L}$ in each state direction:

$$\delta_{v_1}\mathscr{L}(\delta v_1) = \frac{\mathsf{d}}{\mathsf{d}\varepsilon}\mathscr{L}(v_1+\varepsilon\delta v_1, v_2, u_2, p_1, p_2)\bigg|_{\varepsilon=0}, \qquad \delta_{v_2}\mathscr{L}(\delta v_2) = \frac{\mathsf{d}}{\mathsf{d}\varepsilon}\mathscr{L}(v_1, v_2+\varepsilon\delta v_2, u_2, p_1, p_2)\bigg|_{\varepsilon=0}.$$

For the variation in v_1, the relevant terms of $\mathscr{L}$ are

$$\frac{1}{2}\int_Q \gamma_1(v_1-v_1^{\mathrm{tar}})^2\,\mathsf{d}x\,\mathsf{d}t, \qquad -\int_Q p_1\big(\partial_t v_1 - \mu_1\Delta v_1 - v_1(-m+\alpha v_2)\big)\,\mathsf{d}x\,\mathsf{d}t,$$

and

$$-\int_Q p_2\big(\partial_t v_2 - \mu_2\Delta v_2 - v_2(r-\beta v_1) + q u_2 v_2\big)\,\mathsf{d}x\,\mathsf{d}t.$$

Differentiation with respect to v_1 gives

$$\begin{aligned}\delta_{v_1}\mathscr{L}(\delta v_1) = \int_Q \gamma_1(v_1-v_1^{\mathrm{tar}})\,\delta v_1\,\mathsf{d}x\,\mathsf{d}t - \int_Q p_1\,\partial_t(\delta v_1)\,\mathsf{d}x\,\mathsf{d}t + \int_Q p_1\,\mu_1\Delta(\delta v_1)\,\mathsf{d}x\,\mathsf{d}t \\ + \int_Q p_1(-m+\alpha v_2)\,\delta v_1\,\mathsf{d}x\,\mathsf{d}t - \int_Q \beta p_2 v_2\,\delta v_1\,\mathsf{d}x\,\mathsf{d}t.\end{aligned}$$

We integrate the derivative terms by parts. In time,

$$\int_Q p_1\,\partial_t(\delta v_1)\,\mathsf{d}x\,\mathsf{d}t = \left[\int_\Omega p_1(x,t)\,\delta v_1(x,t)\,\mathsf{d}x\right]_{t=0}^{T} - \int_Q \partial_t p_1\,\delta v_1\,\mathsf{d}x\,\mathsf{d}t.$$

Since $\delta v_1(\cdot,0)=0$, only the terminal term remains:

$$\int_Q p_1\,\partial_t(\delta v_1)\,\mathsf{d}x\,\mathsf{d}t = \int_\Omega p_1(x,T)\,\delta v_1(x,T)\,\mathsf{d}x - \int_Q \partial_t p_1\,\delta v_1\,\mathsf{d}x\,\mathsf{d}t.$$

For diffusion, Green's identity gives

$$\int_\Omega p_1\Delta(\delta v_1)\,\mathsf{d}x = \int_\Omega \Delta p_1\,\delta v_1\,\mathsf{d}x + \int_{\partial\Omega}\left(p_1\frac{\partial \delta v_1}{\partial n} - \delta v_1\frac{\partial p_1}{\partial n}\right)\mathsf{d}S.$$

The state variations satisfy $\partial_n\delta v_1 = 0$ because the Neumann condition is fixed. We impose the same homogeneous condition $\partial_n p_1 = 0$ on the adjoint. The two boundary terms in Green's identity then vanish separately, and the diffusion contribution reduces to

$$\int_Q p_1\mu_1\Delta(\delta v_1)\,\mathsf{d}x\,\mathsf{d}t = \int_Q \mu_1\Delta p_1\,\delta v_1\,\mathsf{d}x\,\mathsf{d}t.$$

Collecting all terms,

$$\delta_{v_1}\mathscr{L}(\delta v_1) = \int_Q\Big[\gamma_1(v_1-v_1^{\mathrm{tar}}) + \partial_t p_1 + \mu_1\Delta p_1 + p_1(-m+\alpha v_2) - \beta p_2 v_2\Big]\delta v_1\,\mathsf{d}x\,\mathsf{d}t - \int_\Omega p_1(x,T)\,\delta v_1(x,T)\,\mathsf{d}x.$$

Because δv_1 and $\delta v_1(\cdot,T)$ are arbitrary, their coefficients vanish. Thus $p_1(\cdot,T)=0$ and

$$-\partial_t p_1 - \mu_1\Delta p_1 = \gamma_1(v_1-v_1^{\mathrm{tar}}) + p_1(-m+\alpha v_2) - \beta p_2 v_2.$$

The same calculation for v_2 gives

$$\begin{aligned}\delta_{v_2}\mathscr{L}(\delta v_2) = \int_Q \gamma_2(v_2-v_2^{\mathrm{tar}})\,\delta v_2\,\mathsf{d}x\,\mathsf{d}t - \int_Q p_2\,\partial_t(\delta v_2)\,\mathsf{d}x\,\mathsf{d}t + \int_Q p_2\,\mu_2\Delta(\delta v_2)\,\mathsf{d}x\,\mathsf{d}t \\ + \int_Q p_2\big(r-\beta v_1 - q\,u_2\big)\delta v_2\,\mathsf{d}x\,\mathsf{d}t + \int_Q \alpha p_1 v_1\,\delta v_2\,\mathsf{d}x\,\mathsf{d}t - \int_Q \eta q\,u_2\,\delta v_2\,\mathsf{d}x\,\mathsf{d}t.\end{aligned}$$

After integration by parts, with $\partial_n\delta v_2 = 0$ and $\partial_n p_2 = 0$ on $\partial\Omega\times(0,T)$, we obtain

$$\begin{aligned}\delta_{v_2}\mathscr{L}(\delta v_2) = \int_Q\Big[\gamma_2(v_2-v_2^{\mathrm{tar}}) + \partial_t p_2 + \mu_2\Delta p_2 - \eta q\,u_2 + \alpha p_1 v_1 + p_2(r-\beta v_1 - q\,u_2)\Big]\delta v_2\,\mathsf{d}x\,\mathsf{d}t \\ - \int_\Omega p_2(x,T)\,\delta v_2(x,T)\,\mathsf{d}x.\end{aligned}$$

The arbitrary terminal variation gives $p_2(\cdot,T)=0$, and the coefficient of δv_2 gives

$$-\partial_t p_2 - \mu_2\Delta p_2 = \gamma_2(v_2-v_2^{\mathrm{tar}}) - \eta q\,u_2 + \alpha p_1 v_1 + p_2(r-\beta v_1 - q\,u_2).$$

The complete adjoint system is therefore

$$\begin{cases} -\partial_t p_1 - \mu_1 \Delta p_1 = \gamma_1(v_1 - v_1^{\mathrm{tar}}) + p_1(-m + \alpha v_2) - \beta p_2 v_2, & \text{in } Q, \\ -\partial_t p_2 - \mu_2 \Delta p_2 = \gamma_2(v_2 - v_2^{\mathrm{tar}}) - \eta q\, u_2 + \alpha p_1 v_1 + p_2(r - \beta v_1 - q\, u_2), & \text{in } Q, \\ \dfrac{\partial p_1}{\partial n} = \dfrac{\partial p_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0,T), \\ p_1(\cdot, T) = 0, \qquad p_2(\cdot, T) = 0, & \text{in } \Omega. \end{cases}$$

The terminal conditions $p_1(\cdot,T) = p_2(\cdot,T) = 0$ come from the absence of a terminal cost: the values $v_i(\cdot,T)$ are free, so their terminal variations are arbitrary. The adjoint boundary conditions $\partial_n p_1 = \partial_n p_2 = 0$ cancel the boundary integrals created by Green's formula and match the homogeneous Neumann structure of the state equations.

Let h be an admissible direction and $(z_1, z_2) = S'(u_2)h$. Direct differentiation of the reduced cost gives

$$\begin{aligned} J'(u_2)h = &\int_Q \gamma_1(v_1 - v_1^{\mathrm{tar}}) z_1 + \left[\gamma_2(v_2 - v_2^{\mathrm{tar}}) - \eta q u_2\right] z_2 \,\mathrm{d}x\,\mathrm{d}t \\ &+ \int_Q (\lambda_2 u_2 - \eta q v_2) h \,\mathrm{d}x\,\mathrm{d}t. \end{aligned}$$

Multiply the linearized state equations by p_1, p_2 and integrate by parts in space and time. The terms involving z_1, z_2 cancel by the adjoint equations. Using

$$z_1(\cdot,0) = z_2(\cdot,0) = 0, \qquad p_1(\cdot,T) = p_2(\cdot,T) = 0,$$

and the homogeneous Neumann conditions, we obtain

$$J'(u_2)h = \int_Q \left[\lambda_2 u_2 - \eta q v_2 - q v_2 p_2\right] h \,\mathrm{d}x\,\mathrm{d}t.$$

Thus

$$J'(u_2)h = \int_Q \left[\lambda_2 u_2 - q v_2(\eta + p_2)\right] h \,\mathrm{d}x\,\mathrm{d}t.$$

The relations above are first-order necessary conditions. They do not prove that a stationary control is a global minimizer, because the nonlinear state equation makes the reduced functional nonconvex in general. Different initial controls may therefore lead to different stationary points. Under the box constraint, the derivative is first taken along admissible directions. Since $\mathscr{U}_{\mathrm{ad}}^{\Omega}$ is convex, every segment $u_2^* + s(w - u_2^*)$, $s \in [0,1]$, remains admissible. Let u_2^* be an optimal control, let $(v_1^*, v_2^*) = S(u_2^*)$, and let (p_1^*, p_2^*) solve the corresponding adjoint system. Differentiating along this segment at $s = 0$ gives the variational inequality

$$J'(u_2^*)(w - u_2^*) \geq 0 \qquad \text{for all } w \in \mathscr{U}_{\mathrm{ad}}^{\Omega}.$$

Using $\nabla J(u_2^*) = \lambda_2 u_2^* - q v_2^*(\eta + p_2^*)$, this is

$$\int_Q \left[\lambda_2 u_2^* - q v_2^*(\eta + p_2^*)\right](w - u_2^*) \,\mathrm{d}x\,\mathrm{d}t \geq 0 \qquad \text{for all } w \in \mathscr{U}_{\mathrm{ad}}^{\Omega}.$$

Since $\mathscr{U}_{\mathrm{ad}}^{\Omega}$ is defined pointwise by

$$0 \leq u_2(x,t) \leq U_2^{\max} \quad \text{for a.e. } (x,t) \in Q$$

the variational inequality is equivalent, a.e. in Q, to

$$u_2^*(x,t) = \Pi_{[0, U_2^{\max}]}\left(\frac{q\, v_2^*(x,t)}{\lambda_2}\big(\eta + p_2^*(x,t)\big)\right) \quad \text{for a.e. } (x,t) \in Q.$$

Here $\Pi_{[a,b]}(z) = \min(b, \max(a,z))$. Pointwise in (x,t), the second derivative of the Hamiltonian with respect to the control is $\partial^2_{u_2 u_2}\mathscr{H} = \lambda_2 > 0$. Hence $u_2 \mapsto \mathscr{H}$ is strictly convex and has a unique unconstrained minimizer, namely

$$\frac{q\, v_2^*(x,t)}{\lambda_2}\big(\eta + p_2^*(x,t)\big).$$

The box constraint $0 \leq u_2 \leq U_2^{\max}$ gives the projected formula above.

2.3 Stationary Profiles

We now study stationary profiles, namely time-independent solutions of the controlled spatial system. Such profiles are standard objects in reaction–diffusion theory [18]; they may describe long-time states and may also be prescribed as targets in population control [17]. A stationary profile is a pair (V_1, V_2) depending only on x. Substitution into the parabolic equations gives $\partial_t V_1 = \partial_t V_2 = 0$.

If the control is independent of time, write $u_2(x,t) = U_2(x)$. The stationary equations are

$$\begin{cases} -\mu_1 \Delta V_1 = V_1(-m + \alpha V_2), & \text{in } \Omega, \\ -\mu_2 \Delta V_2 = V_2(r - \beta V_1) - qU_2V_2, & \text{in } \Omega, \\ \dfrac{\partial V_1}{\partial n} = \dfrac{\partial V_2}{\partial n} = 0, & \text{on } \partial\Omega. \end{cases}$$

The stationary control U_2 may be a constant or a function of x. In either case, (V_1, V_2) solves a coupled nonlinear elliptic system [6, 18]; the product V_1V_2 prevents a general closed formula. Homogeneous Neumann conditions admit constant functions because $\partial_n V_i = 0$ automatically. Therefore every equilibrium of the non-spatial controlled system gives a constant stationary profile. When U_2 is constant, the diffusion terms vanish for such a profile and the stationary equations reduce to

$$\begin{cases} V_1(-m + \alpha V_2) = 0, \\ V_2(r - \beta V_1 - qU_2) = 0. \end{cases}$$

The positive constant profile is

$$V_1^* = \frac{r - qU_2}{\beta}, \qquad V_2^* = \frac{m}{\alpha},$$

provided $r - qU_2 > 0$, or equivalently $qU_2 < r$. If $r - qU_2 \le 0$, the positive homogeneous coexistence profile is absent. The elliptic system can still be studied for nonconstant states, especially when U_2 depends on x. Such a profile satisfies the same Neumann conditions but has $\nabla V_i \not\equiv 0$. We write this nonlinear stationary problem as the operator equation

$$F(V_1, V_2) = 0 \quad \text{where} \quad F(V_1, V_2) = \begin{pmatrix} F_1(V_1, V_2) \\ F_2(V_1, V_2) \end{pmatrix},$$

with

$$F_1(V_1, V_2) = -\mu_1 \Delta V_1 - V_1(-m + \alpha V_2), \qquad F_2(V_1, V_2) = -\mu_2 \Delta V_2 - V_2(r - \beta V_1 - qU_2).$$

After expanding the reaction terms,

$$F_1(V_1, V_2) = -\mu_1 \Delta V_1 + mV_1 - \alpha V_1 V_2, \qquad F_2(V_1, V_2) = -\mu_2 \Delta V_2 - rV_2 + \beta V_1 V_2 + qU_2 V_2.$$

The bilinear product V_1V_2 makes the elliptic operator F nonlinear. We use Newton's method [11]. Starting from an initial pair $(V_1^{(0)}, V_2^{(0)})$, each iteration computes a correction by solving the linearized equation

$$DF(V_1^{(k)}, V_2^{(k)}) \begin{pmatrix} \delta V_1^{(k)} \\ \delta V_2^{(k)} \end{pmatrix} = -F(V_1^{(k)}, V_2^{(k)}),$$

and updating

$$V_1^{(k+1)} = V_1^{(k)} + \delta V_1^{(k)}, \qquad V_2^{(k+1)} = V_2^{(k)} + \delta V_2^{(k)}.$$

The Fréchet derivative of the first component is

$$\delta F_1 = -\mu_1 \Delta(\delta V_1) + m\delta V_1 - \alpha V_2 \delta V_1 - \alpha V_1 \delta V_2 = -\mu_1 \Delta(\delta V_1) + (m - \alpha V_2)\delta V_1 - \alpha V_1 \delta V_2.$$

For the second component,

$$\delta F_2 = -\mu_2 \Delta(\delta V_2) + \beta V_2 \delta V_1 + (-r + \beta V_1 + qU_2)\delta V_2.$$

At iteration k, the Newton correction $(\delta V_1^{(k)}, \delta V_2^{(k)})$ therefore solves

$$\begin{cases} -\mu_1 \Delta \delta V_1^{(k)} + \big(m - \alpha V_2^{(k)}\big)\delta V_1^{(k)} - \alpha V_1^{(k)} \delta V_2^{(k)} = -F_1(V_1^{(k)}, V_2^{(k)}), & \text{in } \Omega, \\ \beta V_2^{(k)} \delta V_1^{(k)} - \mu_2 \Delta \delta V_2^{(k)} + \big(-r + \beta V_1^{(k)} + qU_2\big)\delta V_2^{(k)} = -F_2(V_1^{(k)}, V_2^{(k)}), & \text{in } \Omega, \\ \dfrac{\partial \delta V_1^{(k)}}{\partial n} = \dfrac{\partial \delta V_2^{(k)}}{\partial n} = 0, & \text{on } \partial\Omega. \end{cases}$$

The equations are linear in the corrections $\delta V_1^{(k)}$ and $\delta V_2^{(k)}$, but their coefficients are evaluated at the current iterate. After finite-difference discretization, multiplication by $V_i^{(k)}$ becomes a diagonal matrix and diffusion becomes the Neumann matrix Δ_h. If the two grid vectors are stacked into one vector, the Newton equation takes the block form

$$\begin{pmatrix} -\mu_1\Delta_h + mI - \alpha\,\mathrm{diag}(V_2^{(k)}) & -\alpha\,\mathrm{diag}(V_1^{(k)}) \\ \beta\,\mathrm{diag}(V_2^{(k)}) & -\mu_2\Delta_h - rI + \beta\,\mathrm{diag}(V_1^{(k)}) + q\,\mathrm{diag}(U_2) \end{pmatrix} \begin{pmatrix} \delta V_1^{(k)} \\ \delta V_2^{(k)} \end{pmatrix} = -\begin{pmatrix} F_1(V_1^{(k)}, V_2^{(k)}) \\ F_2(V_1^{(k)}, V_2^{(k)}) \end{pmatrix}.$$

The matrix Δ_h is the same reflected Neumann Laplacian used in the reaction–diffusion simulation. Thus its assembly and factorization can be reused. If U_2 is constant and $r - qU_2 > 0$, the positive homogeneous coexistence profile provides the natural Newton initial guess

$$(V_1^*, V_2^*) = \left(\frac{r - qU_2}{\beta}, \frac{m}{\alpha}\right),$$

To test the local behavior of this constant state, we can add a small Neumann-compatible perturbation:

$$V_1^{(0)}(x) = V_1^* + \varepsilon\phi(x), \quad V_2^{(0)}(x) = V_2^* + \varepsilon\psi(x),$$

Here $\varepsilon > 0$ is small and the perturbations satisfy $\partial_n\phi = \partial_n\psi = 0$. If Newton's method returns to (V_1^*, V_2^*), the computation is consistent with local uniqueness of the constant profile for the chosen parameters. If it converges to a different nonconstant pair, this gives numerical evidence for another stationary solution. We can study the first possibility analytically near the homogeneous state. For constant U_2, homogeneous coefficients, and Neumann boundary conditions, decompose a perturbation into Laplacian eigenmodes. The mode-by-mode linearization below shows that no small-amplitude nonconstant stationary branch bifurcates from the positive coexistence profile. We therefore linearize the stationary system at

$$V_1^* = \frac{r - qU_2}{\beta}, \quad V_2^* = \frac{m}{\alpha}.$$

Expand the perturbations in Neumann eigenfunctions [6, 18]. Let

$$-\Delta\phi_k = \lambda_k\phi_k, \qquad \frac{\partial\phi_k}{\partial n} = 0 \quad \text{on } \partial\Omega.$$

The mode $\lambda_0 = 0$ is constant; every $\lambda_k > 0$ is spatially varying. For a fixed mode, set

$$V_1 = V_1^* + \xi_1\phi_k, \quad V_2 = V_2^* + \xi_2\phi_k.$$

The linearized coefficients satisfy

$$\begin{pmatrix} \mu_1\lambda_k & -\alpha V_1^* \\ \beta V_2^* & \mu_2\lambda_k \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = 0.$$

A nonzero vector (ξ_1, ξ_2) exists only when the determinant is zero:

$$\mu_1\mu_2\lambda_k^2 + \alpha\beta V_1^* V_2^* = 0.$$

In the coexistence regime, $V_1^*, V_2^* > 0$ and $\mu_1, \mu_2, \alpha, \beta > 0$. Hence

$$\mu_1\mu_2\lambda_k^2 + \alpha\beta V_1^* V_2^* > 0$$

for every Neumann mode k. The determinant is strictly positive for $k = 0$ and for all $k \geq 1$, so the linearized stationary operator has no kernel on any eigenspace of the Neumann Laplacian. In particular, the positive constant profile does not generate a small-amplitude nonconstant branch through this linearized mechanism. This is only a local statement near (V_1^*, V_2^*): it does not exclude a nonconstant solution lying far from the homogeneous state. Spatial profiles may still appear if the control is heterogeneous, $U_2 = U_2(x)$, if one of the coefficients depends on x, if another boundary condition is imposed, or if the reaction law is changed. With constant coefficients, constant control, and homogeneous Neumann conditions, the natural stationary candidate is therefore the constant coexistence profile whenever $r - qU_2 > 0$. To give a concrete nonconstant example without changing the biological reactions, we now choose a time-independent but spatially varying control $U_2(x)$ and consider the system

$$\begin{cases} -\mu_1\Delta V_1 = V_1(-m + \alpha V_2), & \text{in } \Omega, \\ -\mu_2\Delta V_2 = V_2(r - \beta V_1) - qU_2(x)V_2, & \text{in } \Omega, \\ \dfrac{\partial V_1}{\partial n} = \dfrac{\partial V_2}{\partial n} = 0, & \text{on } \partial\Omega. \end{cases}$$

We compute this stationary state by integrating the associated parabolic system over a long time interval. The iteration is stopped when the difference between two successive states satisfies $\|v^{n+1} - v^n\|_\infty < \varepsilon_{\text{stat}}$. Because the stationary control $U_2(x)$ is not constant, the limiting reaction balance varies with x and the equilibrium densities are nonconstant. Figure 8 shows the result.

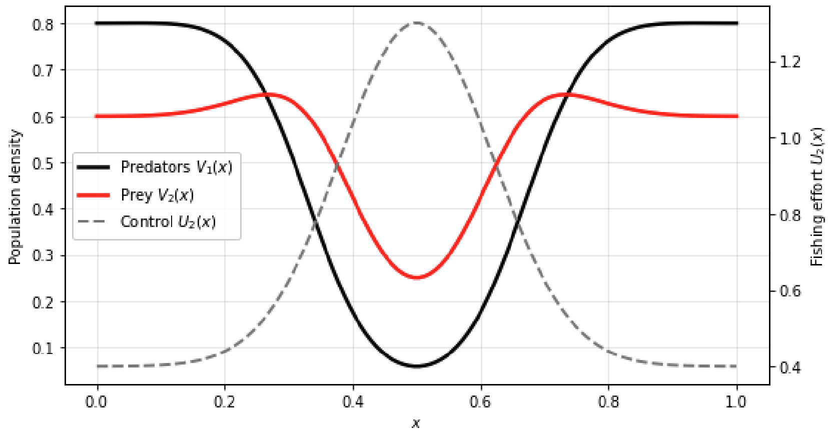

Figure 8: Nonconstant stationary densities generated by a spatially heterogeneous control $U_2(x)$.

2.4 Numerical Computation of the Optimal Control

We now discretize and solve the optimality system. The unknowns are the states (v_1, v_2), the adjoints (p_1, p_2), and the control u_2. A forward–backward sweep performs, at each iteration, one forward state solve, one backward adjoint solve, and one pointwise control update. Forward–backward sweeps are classical for optimality systems derived from Pontryagin's principle, and their convergence has been studied in [13]. We first present a semi-implicit first-order version in which diffusion is solved implicitly and the local terms remain explicit. This gives a transparent and inexpensive reference algorithm. We then explain the stability and convergence limits of this fixed-point iteration and compare it with two reduced optimization methods, projected gradient and L-BFGS-B. In all three cases, special attention is given to the discrete adjoint: it is built from the fully discrete state recurrence so that the computed vector is the gradient of the discrete cost actually minimized by the code.

A first-order semi-implicit forward–backward sweep method

We use a first-order semi-implicit time discretization [10, 16]. Diffusion is evaluated at the new time level, while the nonlinear reactions and the control term are evaluated at the old time level. Each forward step therefore requires two linear elliptic solves. This choice removes the severe diffusion restriction of a fully explicit method and keeps the implementation simple, but it does not make the nonlinear part implicit. We follow a discretize–then–optimize strategy [12, 20]. First, the state recurrence and the quadrature defining the cost are fixed. Next, the discrete adjoint is obtained by transposing the linearized discrete state map. The resulting vector is exactly the gradient of the finite-dimensional reduced objective. This point is important: solving a time discretization of the continuous adjoint can produce a nearby formula, but it need not give the exact derivative of the chosen discrete cost. Let $t_n = n\Delta t$, $n = 0, \ldots, N$, with $\Delta t = T/N$. We approximate u_2 by a control that is constant on each interval $[t_n, t_{n+1})$ and denote its value by $u_{2,n}$, $n = 0, \ldots, N-1$. Let Δ_h be the finite-difference Laplacian with homogeneous Neumann conditions. On the uniform grid, it is self-adjoint for the discrete L^2 inner product used in the cost. Consequently, the adjoint diffusion solve uses the adjoint of $I - \Delta t\, \mu_i \Delta_h$, which has the same form as the forward matrix in this inner product.

The complete first-order forward–backward sweep is as follows.

Input: final time T, number of time steps N, time step $\Delta t = T/N$, domain $\Omega \subseteq \mathbf{R}^d$, spatial mesh, initial states $v_{1,0} = v_1^0$ and $v_{2,0} = v_2^0$, control bound $U_2^{\max}$, model and cost parameters, target profiles, relaxation $0 < \theta \leq 1$, tolerance $\varepsilon > 0$, and maximum iteration count $K_{\max}$.

Initialization:

- set $t_n = n\Delta t$, $n = 0, \ldots, N$,
- choose the spatial grid on Ω,
- choose $u_{2,n}^{(0)}(x) \in [0, U_2^{\max}]$, $n = 0, \ldots, N-1$, for example $u_{2,n}^{(0)} \equiv 0$ or a constant,

- set $k=0$ and $E^{(0)}=+\infty$.

While $k<K_{\max}$ and $E^{(k)}\geq\varepsilon$:

- **Step 1: forward solution of the state system.**

  Starting from $v_{1,0}^{(k)}=v_1^0$ and $v_{2,0}^{(k)}=v_2^0$, compute $n=0,\dots,N-1$:

$$\begin{cases}(I-\Delta t\,\mu_1\Delta_h)\,v_{1,n+1}^{(k)}=v_{1,n}^{(k)}+\Delta t\,v_{1,n}^{(k)}\big(-m+\alpha v_{2,n}^{(k)}\big),\\ (I-\Delta t\,\mu_2\Delta_h)\,v_{2,n+1}^{(k)}=v_{2,n}^{(k)}+\Delta t\left[v_{2,n}^{(k)}\big(r-\beta v_{1,n}^{(k)}\big)-q\,u_{2,n}^{(k)}v_{2,n}^{(k)}\right].\end{cases}$$

  Each forward step therefore consists of two linear elliptic solves.

- **Step 2: backward solution of the discrete adjoint system.**
  The discrete cost is evaluated at t_n, $n=0,\dots,N-1$. Since there is no terminal term, set $p_{1,N}^{(k)}(x)=p_{2,N}^{(k)}(x)=0$.
  For $n=N-1,\dots,0$, solve

$$\begin{cases}\dfrac{p_{1,n}^{(k)}(x)-p_{1,n+1}^{(k)}(x)}{\Delta t}-\mu_1\Delta_h p_{1,n}^{(k)}(x)=\gamma_1\big(v_{1,n}^{(k)}(x)-v_{1,n}^{\mathrm{tar}}(x)\big)+\big(-m+\alpha v_{2,n}^{(k)}(x)\big)p_{1,n+1}^{(k)}(x)\\ \qquad\qquad -\beta v_{2,n}^{(k)}(x)p_{2,n+1}^{(k)}(x), & \text{in }\Omega,\\ \dfrac{p_{2,n}^{(k)}(x)-p_{2,n+1}^{(k)}(x)}{\Delta t}-\mu_2\Delta_h p_{2,n}^{(k)}(x)=\gamma_2\big(v_{2,n}^{(k)}(x)-v_{2,n}^{\mathrm{tar}}(x)\big)-\eta q\,u_{2,n}^{(k)}(x)+\alpha v_{1,n}^{(k)}(x)p_{1,n+1}^{(k)}(x)\\ \qquad\qquad +\big(r-\beta v_{1,n}^{(k)}(x)-q\,u_{2,n}^{(k)}(x)\big)p_{2,n+1}^{(k)}(x), & \text{in }\Omega,\\ \dfrac{\partial p_{1,n}^{(k)}}{\partial n}=\dfrac{\partial p_{2,n}^{(k)}}{\partial n}=0, & \text{on }\partial\Omega.\end{cases}$$

  Step 3: control update.
  For $n=0,\dots,N-1$ and at every spatial node, compute the unprojected control

$$u_{r,2,n}^{(k+1)}(x)=\frac{q\,v_{2,n}^{(k)}(x)}{\lambda_2}\Big(\eta+p_{2,n+1}^{(k)}(x)\Big).$$

  Then apply relaxation and projection:

$$u_{2,n}^{(k+1)}(x)=\Pi_{[0,U_2^{\max}]}\Big(\theta u_{r,2,n}^{(k+1)}(x)+(1-\theta)u_{2,n}^{(k)}(x)\Big),\qquad 0<\theta\leq 1.$$

  The parameter $\theta\in(0,1]$ mixes the new pointwise value with the previous iterate; its numerical effect is studied below. The projection is defined by

$$\Pi_{[0,U_2^{\max}]}(z)=\min\left(U_2^{\max},\max(0,z)\right).$$

  Thus $u_{2,n}^{(k+1)}(x)\in[0,U_2^{\max}]$ at every mesh point.

  Step 4: convergence test.
  Compute

$$E^{(k+1)}=\max_{0\leq n\leq N-1}\left\|u_{2,n}^{(k+1)}-u_{2,n}^{(k)}\right\|_{L^\infty(\Omega)}.$$

  On the grid, $E^{(k+1)}$ is computed with the discrete maximum norm.
  If $E^{(k+1)}<\varepsilon$, stop.
  Otherwise set $k\leftarrow k+1$ and repeat.

Output: discrete state and adjoint trajectories $v_{1,n}^{\mathrm{opt}},v_{2,n}^{\mathrm{opt}},p_{1,n}^{\mathrm{opt}},p_{2,n}^{\mathrm{opt}}$, $n=0,\dots,N$, and the discrete control $u_{2,n}^{\mathrm{opt}}$, $n=0,\dots,N-1$.

Limitations and possible improvements

This scheme is simple and inexpensive, but it has clear limits. Its temporal order is one: diffusion uses implicit Euler and the local terms use explicit Euler. As a result, the computed control may change when Δt changes, especially near sharp temporal variations. Implicit diffusion removes the main diffusion restriction, but the explicit Lotka–Volterra terms can still create oscillations or negative densities when Δt is large, the populations are high, or the coupling coefficients are strong. A positivity projection can remove negative nodal values, but whenever it is active it changes the formal scheme and its local error. The forward–backward sweep is also a fixed-point iteration. Its convergence requires a contraction estimate and may fail for large T, strong state–adjoint coupling, or a control update that moves too far in one step. Relaxation can damp oscillations, but the condition $K_h < 1$ derived below remains the key one. In addition, the discrete cost j_h need not decrease at every sweep. A converged fixed point satisfies the first-order discrete optimality system, but it may be a local stationary point of the nonconvex problem. These limits motivate descent methods for finite-dimensional box-constrained optimization, such as projected gradient and limited-memory quasi-Newton methods [4, 5, 22].

Reduced formulation and projected-gradient method

A second strategy treats the same fully discrete problem as a finite-dimensional optimization problem and minimizes the reduced cost directly. Let $S_h : u_h \mapsto v_h$ denote the discrete control-to-state map, where

$$u_h := \big(u_{2,n}(x_i)\big)_{\substack{0\le n\le N-1\\ 0\le i\le M}} \in \mathbf{R}^{N(M+1)}$$

collects the $N(M+1)$ control values and $v_h = S_h(u_h)$ is the full state trajectory. Define

$$j_h(u_h) := J_h(S_h(u_h), u_h).$$

The finite-dimensional problem is therefore $\min_{u_h\in\mathscr{U}_{\mathrm{ad},h}} j_h(u_h)$, where

$$\mathscr{U}_{\mathrm{ad},h} := \left\{ u_h = \big(u_{2,n}(x_i)\big)_{\substack{0\le n\le N-1\\ 0\le i\le M}} \in \mathbf{R}^{N(M+1)} \;\middle|\; 0 \le u_{2,n}(x_i) \le U_2^{\max} \text{ for all } n, i \right\}.$$

The reduced formulation removes the state from the list of optimization variables. For a trial u_h, the trajectory $v_h = S_h(u_h)$ is found by the forward recurrence, and one discrete adjoint solve provides all components of the gradient. The cost of the gradient is therefore essentially independent of the number of control components. For the first-order adjoint-consistent scheme defined above, the gradient component associated with $u_{2,n}$ is

$$\nabla j_h(u_{2,n}) = \lambda_2 u_{2,n} - \eta q v_{2,n} - q v_{2,n} p_{2,n+1}.$$

The box-constrained first-order condition is measured by the projected-gradient residual

$$R_h(u_h) = u_h - \Pi_{\mathscr{U}_{\mathrm{ad},h}}\big(u_h - \nabla j_h(u_h)\big).$$

The equation $R_h(u_h) = 0$ is equivalent to the projected first-order optimality condition for the finite-dimensional box-constrained problem. It also provides a natural stopping criterion when some components are active at 0 or $U_2^{\max}$. Following the standard projected-gradient method [4], we update the control by

$$u_h^{(k+1)} = \Pi_{\mathscr{U}_{\mathrm{ad},h}}\left(u_h^{(k)} - \rho_k \nabla j_h(u_h^{(k)})\right),$$

where $\rho_k > 0$ is the step size. The projection is componentwise, so every iterate remains in $\mathscr{U}_{\mathrm{ad},h}$ and satisfies $0 \le u_{2,n}(x_i) \le U_2^{\max}$. We choose ρ_k with an Armijo backtracking rule [4]. Starting from $u_h^{(0)} \in \mathscr{U}_{\mathrm{ad},h}$, one iteration first solves the state equation forward, then the discrete adjoint backward, and finally computes $g_h^{(k)} = \nabla j_h(u_h^{(k)})$. We begin with a trial value $\rho_0 > 0$ and form the projected displacement. If the cost decrease is too small, we replace ρ_k by $\tau\rho_k$, with a fixed $\tau \in (0,1)$, and test again. This procedure continues until the projected direction

$$d_k = \Pi_{\mathscr{U}_{\mathrm{ad},h}}\left(u_h^{(k)} - \rho_k \nabla j_h(u_h^{(k)})\right) - u_h^{(k)},$$

satisfies the Armijo inequality

$$j_h(u_h^{(k)} + d_k) \le j_h(u_h^{(k)}) + \sigma \left\langle \nabla j_h(u_h^{(k)}), d_k \right\rangle_h,$$

where $\sigma \in (0,1)$ and $\langle\cdot,\cdot\rangle_h$ is the discrete inner product used to represent the gradient. A common alternative is the sufficient-decrease estimate

$$j_h(u_h^{(k)} + d_k) \le j_h(u_h^{(k)}) - c\,\frac{\|d_k\|_h^2}{\rho_k},$$

for some $c > 0$. Once the inequality is satisfied, we accept $u_h^{(k+1)} = u_h^{(k)} + d_k$. The algorithm stops when the projected residual is small, for example when $\|R_h(u_h^{(k)})\|_h < \varepsilon$. This residual is adapted to the box constraints: it vanishes exactly when the projected first-order condition holds. The Armijo rule makes the accepted step a descent step for the discrete objective, which is a stronger numerical safeguard than the direct FBS fixed-point update. Each outer iteration still uses one forward state solve and one backward adjoint solve. Backtracking may require several additional evaluations of j_h, and therefore several extra forward solves, but it avoids accepting a step that increases the discrete cost.

Quasi-Newton reduced optimization

A third option is the bound-constrained quasi-Newton method L-BFGS-B [5, 22]. After discretization, the distributed control becomes a vector $u_h \in \mathbf{R}^{N(M+1)}$ with componentwise constraints $0 \le u_h \le U_2^{\max}$. The state variables are still eliminated through S_h, so the only optimization unknown is u_h and the reduced objective is

$$j_h(u_h) = J_h(S_h(u_h), u_h)$$

The value $j_h(u_h)$ is obtained from one forward state solve, and $\nabla j_h(u_h)$ is obtained from one backward discrete adjoint solve. L-BFGS-B then builds a limited-memory approximation of the inverse reduced Hessian. More precisely, it stores only a small number of differences $s_k = u_h^{(k+1)} - u_h^{(k)}$ and $y_k = \nabla j_h(u_h^{(k+1)}) - \nabla j_h(u_h^{(k)})$. These pairs contain local curvature information and define a quasi-Newton search direction without forming the full Hessian in $\mathbf{R}^{N(M+1)\times N(M+1)}$. The method also identifies active components at the lower and upper bounds. Its update is more involved than one projected-gradient step, but the curvature model often reduces the number of state and adjoint solves needed to reach a small projected residual.

Comparison of the methods

The three methods use the same state equation and the same adjoint-consistent gradient. FBS is a low-cost fixed-point baseline; projected gradient is a descent method with a line search; L-BFGS-B is a box-constrained quasi-Newton method that also uses curvature information.

Numerical comparison of the optimization methods

For the numerical benchmark [2], we take $\Omega = (0,1)$ and $T = 4$. The interval is divided into $M = 80$ subintervals, hence $M+1$ nodes, and the time interval is divided into $N = 120$ steps. We set $m = 1$, $r = 1.2$, $\alpha = \beta = q = 1$, and $\mu_1 = \mu_2 = 0.002$. Homogeneous Neumann conditions are used at both endpoints. The localized initial densities are

$$v_1^0(x) = 1.3\exp\left(-\frac{(x-0.30)^2}{2\cdot 0.10^2}\right) + 0.05, \quad v_2^0(x) = 1.4\exp\left(-\frac{(x-0.70)^2}{2\cdot 0.10^2}\right) + 0.05.$$

The targets are time-independent perturbations of the uncontrolled coexistence state $(r/\beta, m/\alpha)$:

$$v_1^{\text{tar}}(x) = \frac{r}{\beta} + 0.15\cos(2\pi x), \qquad v_2^{\text{tar}}(x) = \frac{m}{\alpha} - 0.15\cos(2\pi x).$$

The admissible set is defined by $0 \le u_2(x,t) \le U_2^{\max}$ with $U_2^{\max} = 2$. We choose $\gamma_1 = \gamma_2 = 0.5$, $\lambda_2 = 0.05$, and $\eta = 0.8$. Every algorithm starts from the same admissible control $u_2^{(0)} \equiv 0$, uses at most 100 outer iterations, and stops when the corresponding first-order residual is below 10^{-6}. For the basic forward–backward sweep, the default relaxation is $\theta = 0.5$. Projected gradient uses the Armijo line search described above, and L-BFGS-B is called with the same lower and upper bounds. The spatial grid, the time grid, the state solver, the cost quadrature, and the discrete adjoint are identical in all three tests. The comparison therefore concerns the optimization updates, not different discretizations. We report the final value of $j_h(u_h)$, the iteration count, the CPU time, and the projected-gradient residual

$$\left\| u_h - \Pi_{\mathscr{U}_{\text{ad},h}}(u_h - \nabla j_h(u_h)) \right\|_\infty.$$

This residual measures the violation of the projected first-order condition under the box constraints. It is zero at a discrete stationary point and remains meaningful when some components of the control lie on 0 or $U_2^{\max}$. The discrete objective contains the reward term $-\eta q u_2 v_2$, so a negative final value is possible and simply means that the weighted yield exceeds the positive tracking and effort contributions. Table 1 compares FBS with $\theta = 0.5$, projected gradient with Armijo backtracking, and reduced L-BFGS-B.

Method	$j_h(u_h)$	Iterations	CPU time (s)	Residual
FBS, $\theta = 0.5$	2.696596	100	0.688705	6.91×10^{-3}
Projected gradient	1.913330	100	1.423093	3.68×10^{-3}
L-BFGS-B	-1.552726	38	0.616277	8.53×10^{-7}

Table 1: Comparison of the optimization methods.

L-BFGS-B gives the lowest cost and residual and stops after the fewest iterations. Since the FBS result depends on θ, we also use

$$\theta \in \{1, 0.75, 0.5, 0.25, 0.1, 0.05\}.$$

The relaxation parameter θ controls how much of the new pointwise minimizer is inserted at each sweep. A large value moves the control quickly but can amplify the oscillations of the coupled state–adjoint fixed point. A smaller value damps these oscillations, while a very small value may require many iterations because the old control receives weight $1-\theta$. Relaxation alone cannot turn a map with $K_h \geq 1$ into a contraction, as the theorem in Section 2.5 makes explicit. To measure the practical effect, we repeat the same benchmark for all listed values of θ and record the discrete cost $j_h(u_h^k)$, the RMS norm of the successive updates, and the mean control. The plots are generated with the Python code in the GitHub repository [2].

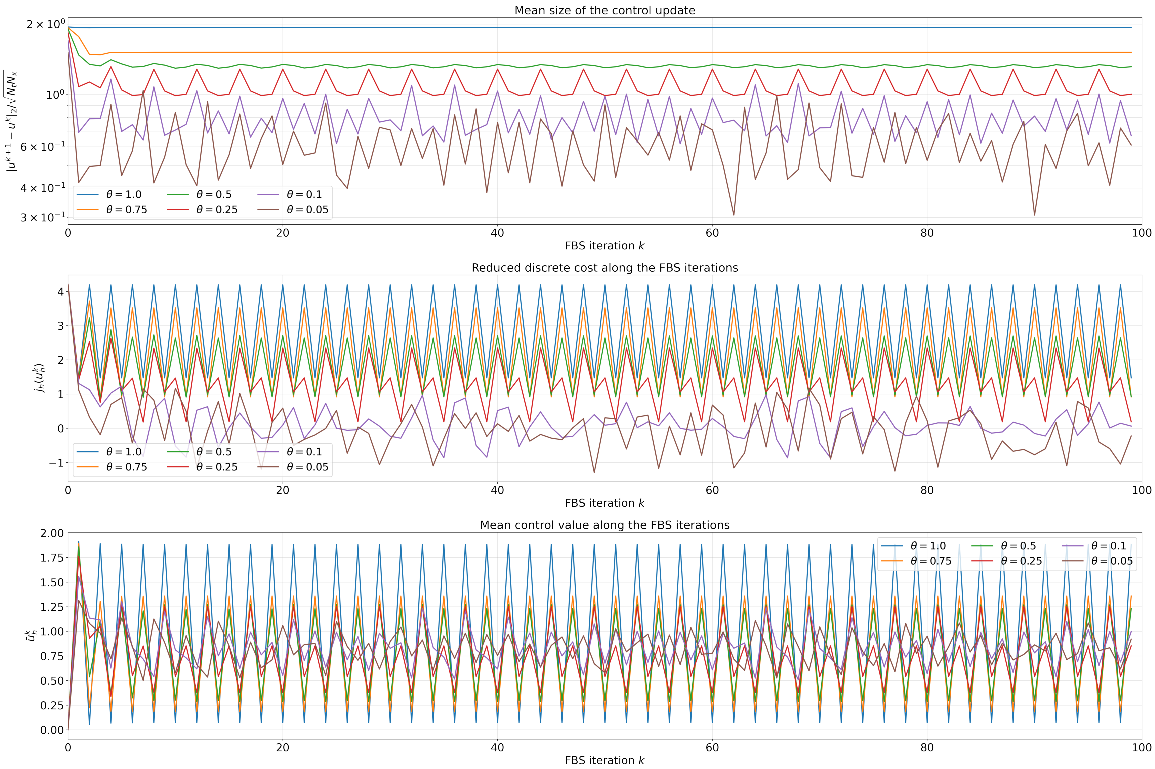

Figure 9: Influence of the relaxation parameter θ on the forward–backward sweep. The top panel gives the RMS norm of the update $u_h^{k+1} - u_h^k$, the middle panel gives the discrete cost $j_h(u_h^k)$, and the bottom panel gives the mean control. Decreasing θ damps the oscillations of the fixed-point iteration. A very small value also reduces the motion of the iterates and can therefore slow convergence.

Figure 9 shows that decreasing θ reduces the amplitude of the successive-control oscillations and changes the cost trajectory. It also shows that small updates can slow the motion toward a fixed point. Actual convergence still depends on the contractivity of the fully discrete map. In Table 2, $\overline{u}_h$ denotes the average of the control over all spatial nodes and time intervals.

θ	$j_h(u_h)$	Iterations	CPU time (s)	Residual	$\overline{u}_h$
1.00	4.189759	100	0.700941	6.811×10^{-3}	0.072166
0.75	3.517737	100	0.690913	6.828×10^{-3}	0.183496
0.50	2.696596	100	0.688705	6.908×10^{-3}	0.286550
0.25	2.345476	100	0.689804	7.265×10^{-3}	0.377919
0.10	0.435958	100	0.689668	5.857×10^{-3}	0.650697
0.05	-0.330496	100	0.695693	9.18×10^{-4}	0.941266

Table 2: Influence of the relaxation parameter θ on the forward–backward sweep method.

Among the tested values, $\theta = 0.05$ gives the best FBS iterate after 100 sweeps: it has the smallest FBS cost and the smallest FBS residual in Table 2. It still remains above the L-BFGS-B result, which reaches a lower cost and a residual below 10^{-6} after 38 iterations. Projected gradient also reaches the prescribed iteration limit. For this parameter set, L-BFGS-B is the most effective solver in terms of cost, stationarity, and iteration count, while the forward–backward sweep remains useful as a simple reference implementation.

Interpretation in terms of fishing.

The value $u_2(x,t)$ identifies where and when fishing is useful. At points where $u_2 = U_2^{\max}$, the marginal gain $qv_2(\eta + p_2)$ is at least the marginal effort cost $\lambda_2 U_2^{\max}$; the upper bound is therefore active.

2.5 Fully Discrete Conditional Convergence of the Forward–Backward Sweep Method

We finish with a conditional convergence theorem for the fully discrete forward–backward sweep, following the contraction estimates in [13]. The spatial mesh, the time step, and the final time are fixed in this section. The result concerns convergence of the resulting finite-dimensional map; it does not claim mesh-independent constants or convergence to a continuous optimal control as $h, \Delta t \to 0$. Let Δ_h be the discrete Laplacian with homogeneous Neumann conditions. We assume that Δ_h is nonpositive and self-adjoint for the discrete inner product used to define the adjoint. For the reflected ghost-point stencil, this property holds for the trapezoidal inner product, and the same statement can be expressed through a diagonal similarity transformation of the matrix. With $\mu_i \geq 0$, $i = 1, 2$, define

$$A_i := I - \Delta t\, \mu_i \Delta_h.$$

Let A_i^* denote the adjoint of A_i for this discrete inner product. We assume that both A_i and A_i^* are invertible and satisfy the maximum-norm stability bounds

$$\|A_i^{-1}\|_\infty \leq 1, \qquad \|(A_i^*)^{-1}\|_\infty \leq 1, \qquad i = 1, 2.$$

These estimates hold, for example, for M-matrices with a discrete maximum principle [10, 15, 16]. In the Euclidean inner product, $A_i^* = A_i^T$. For $w_h = (w_n)_{n=0}^N$, set

$$\|w_h\|_{\infty,h} := \max_{0 \leq n \leq N} \|w_n\|_{\ell^\infty}.$$

For $u_{2,h} = (u_{2,n})_{n=0}^{N-1}$, set

$$\|u_{2,h}\|_{\infty,h} := \max_{0 \leq n \leq N-1} \|u_{2,n}\|_{\ell^\infty}.$$

For $w_h = (w_{1,h}, w_{2,h})$, use the product norm

$$\|w_h\|_{\infty,h} := \|w_{1,h}\|_{\infty,h} + \|w_{2,h}\|_{\infty,h}.$$

The discrete admissible set is

$$\mathscr{U}_{\mathrm{ad},h} = \left\{ u_{2,h} = (u_{2,n})_{0 \leq n \leq N-1} \;:\; 0 \leq u_{2,n}(x_j) \leq U_2^{\max} \text{ for all } n \text{ and all mesh points } x_j \right\}.$$

The finite-dimensional set $\mathscr{U}_{\mathrm{ad},h}$ is closed, hence complete. For $u_{2,h} \in \mathscr{U}_{\mathrm{ad},h}$, the state recursion is

$$\begin{cases} A_1 v_{1,n+1} = v_{1,n} + \Delta t\, v_{1,n}(-m + \alpha v_{2,n}), \\ A_2 v_{2,n+1} = v_{2,n} + \Delta t \left[v_{2,n}(r - \beta v_{1,n}) - q u_{2,n} v_{2,n} \right]. \end{cases}$$

The adjoint is then computed backward from the zero terminal data $p_{1,N} = p_{2,N} = 0$. For $n = N-1, \ldots, 0$, its recursion is

$$\begin{cases} A_1^* p_{1,n} = p_{1,n+1} + \Delta t\, G_{1,n}, \\ A_2^* p_{2,n} = p_{2,n+1} + \Delta t\, G_{2,n}. \end{cases}$$

where

$$G_{1,n} := \gamma_1(v_{1,n} - v_{1,n}^{\mathrm{tar}}) + p_{1,n+1}(-m + \alpha v_{2,n}) - \beta p_{2,n+1} v_{2,n},$$

and

$$G_{2,n} := \gamma_2(v_{2,n} - v_{2,n}^{\mathrm{tar}}) - \eta q\, u_{2,n} + \alpha p_{1,n+1} v_{1,n} + p_{2,n+1}(r - \beta v_{1,n} - q u_{2,n}).$$

For the weighted discrete inner product naturally associated with the Neumann stencil, $A_i^* = A_i$. The same factorizations can then be used in the forward and backward solves, which is useful in practice. If one uses the standard Euclidean inner product with the nonsymmetric ghost-point matrix, the exact discrete adjoint uses A_i^T instead. This distinction is part of the discretize–then–optimize construction: the transpose is determined by the inner product in which the gradient is represented. Using the projected optimality condition from the previous subsection, define the relaxed control map by

$$\Phi_h(u_{2,h})_n = \Pi_{[0,U_2^{\max}]}\left((1-\theta)u_{2,n} + \theta\frac{qv_{2,n}}{\lambda_2}(\eta + p_{2,n+1})\right), \qquad 0 < \theta \le 1,$$

for $n = 0, \dots, N-1$. The projection guarantees $\Phi_h(u_{2,h}) \in \mathscr{U}_{\mathrm{ad},h}$, so Φ_h is a self-map of the complete admissible set. The relaxed FBS iteration is $u_{2,h}^{(k+1)} = \Phi_h(u_{2,h}^{(k)})$. If $u_{2,h} = \Phi_h(u_{2,h})$, then the relaxation term cancels and the same projected optimality condition is obtained for every $\theta \in (0, 1]$.

Theorem 2.4 (Fully discrete conditional convergence)

Assume that the state and adjoint recursions, with the matrices A_i^*, are defined for every $u_{2,h} \in \mathscr{U}_{\mathrm{ad},h}$. Let $0 < \theta \le 1$, and suppose that all generated trajectories satisfy the uniform bounds

$$\|v_{1,h}\|_{\infty,h} + \|v_{2,h}\|_{\infty,h} \le M_v, \qquad \|p_{1,h}\|_{\infty,h} + \|p_{2,h}\|_{\infty,h} \le M_p,$$

Assume also that the discrete reactions and the adjoint source terms are Lipschitz on the bounded set determined by M_v, M_p, and $U_2^{\max}$. This assumption is automatic here because every local term is a polynomial in the bounded variables. The forward recurrence then depends Lipschitz-continuously on the control, and the backward recurrence depends Lipschitz-continuously on the state and control. Thus there are constants $C_{S,h}, C_{P,h} > 0$, independent of the particular pair $u_{2,h}, \widehat{u}_{2,h} \in \mathscr{U}_{\mathrm{ad},h}$, such that

$$\|v_h - \widehat{v}_h\|_{\infty,h} \le C_{S,h}\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}, \quad \|p_h - \widehat{p}_h\|_{\infty,h} \le C_{P,h}\left(\|v_h - \widehat{v}_h\|_{\infty,h} + \|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}\right),$$

where $v_h, \widehat{v}_h$ are generated by $u_{2,h}, \widehat{u}_{2,h}$, and $p_h, \widehat{p}_h$ are their adjoints. The constants $C_{S,h}$ and $C_{P,h}$ may depend on the spatial discretization, Δt, the final time T, the biological and cost parameters, and the bounds M_v, M_p, but they are uniform with respect to the two admissible controls being compared. Define

$$K_h := \frac{q}{\lambda_2}\left[(\eta + M_p)C_{S,h} + M_v C_{P,h}(1 + C_{S,h})\right].$$

Then

$$\|\Phi_h(u_{2,h}) - \Phi_h(\widehat{u}_{2,h})\|_{\infty,h} \le \kappa_h\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}, \qquad \kappa_h := (1-\theta) + \theta K_h.$$

Since $0 < \theta \le 1$, $\kappa_h < 1$ if and only if $K_h < 1$. Under this condition, Φ_h is a contraction on $\mathscr{U}_{\mathrm{ad},h}$. Banach's fixed-point theorem then gives a unique fixed point $u_{2,h}^{\infty} \in \mathscr{U}_{\mathrm{ad},h}$. For every initial control $u_{2,h}^{(0)}$, the iterates $u_{2,h}^{(k+1)} = \Phi_h(u_{2,h}^{(k)})$ converge to $u_{2,h}^{\infty}$ in the discrete maximum norm. The Lipschitz estimates for the forward and backward recursions then give convergence of the associated states and adjoints. Passing to the limit in all three updates shows that the limiting quintuple satisfies the fully discrete optimality system.

Proof. Choose two controls $u_{2,h}, \widehat{u}_{2,h} \in \mathscr{U}_{\mathrm{ad},h}$. Let $(v_{1,h}, v_{2,h})$ and $(\widehat{v}_{1,h}, \widehat{v}_{2,h})$ be the associated state trajectories, and let $(p_{1,h}, p_{2,h})$ and $(\widehat{p}_{1,h}, \widehat{p}_{2,h})$ be their discrete adjoints. For $n = 0, \dots, N-1$, set

$$\delta u_{2,n} := u_{2,n} - \widehat{u}_{2,n}, \qquad \delta v_{i,n} := v_{i,n} - \widehat{v}_{i,n}, \qquad \delta p_{i,n} := p_{i,n} - \widehat{p}_{i,n}.$$

Step 1: Lipschitz dependence of the discrete state on the control.

Subtracting the two state recursions gives

$$\begin{cases} A_1\delta v_{1,n+1} = \delta v_{1,n} + \Delta t\left[v_{1,n}(-m + \alpha v_{2,n}) - \widehat{v}_{1,n}(-m + \alpha\widehat{v}_{2,n})\right], \\ A_2\delta v_{2,n+1} = \delta v_{2,n} + \Delta t\left[R_2(v_{1,n}, v_{2,n}, u_{2,n}) - R_2(\widehat{v}_{1,n}, \widehat{v}_{2,n}, \widehat{u}_{2,n})\right], \end{cases}$$

where $R_2(v_1, v_2, u_2) := v_2(r - \beta v_1) - qu_2v_2$.

On the bounded set fixed by M_v and $U_2^{\max}$, the reaction terms are Lipschitz. Hence, for some $L_S > 0$,

$$\|\delta v_{1,n+1}\|_{\ell^\infty} + \|\delta v_{2,n+1}\|_{\ell^\infty} \le (1 + \Delta t L_S)\left(\|\delta v_{1,n}\|_{\ell^\infty} + \|\delta v_{2,n}\|_{\ell^\infty}\right) + \Delta t\, qM_v\|\delta u_{2,n}\|_{\ell^\infty}.$$

Using $\|A_i^{-1}\|_\infty \le 1$, define

$$E_n^v := \|\delta v_{1,n}\|_{\ell^\infty} + \|\delta v_{2,n}\|_{\ell^\infty}.$$

Since both states have the same initial data, $E_0^v = 0$, and

$$E_{n+1}^v \le (1 + \Delta t L_S)E_n^v + \Delta t\, qM_v\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}.$$

The discrete Gronwall lemma [10, 16] gives

$$E_n^v \le qM_vTe^{L_ST}\|u_{2,h}-\widehat{u}_{2,h}\|_{\infty,h}.$$

Therefore

$$\|v_h-\widehat{v}_h\|_{\infty,h} \le C_{S,h}\|u_{2,h}-\widehat{u}_{2,h}\|_{\infty,h},$$

with, for example, $C_{S,h}=qM_vTe^{L_ST}$.

Step 2: Lipschitz dependence of the discrete adjoint on the state and control.

Both adjoint trajectories start from the same terminal value: $p_{1,N}=\widehat{p}_{1,N}=0$ and $p_{2,N}=\widehat{p}_{2,N}=0$. Subtracting the two backward recursions gives

$$\begin{cases} A_1^*\delta p_{1,n} = \delta p_{1,n+1}+\Delta t\,\delta G_{1,n}, \\ A_2^*\delta p_{2,n} = \delta p_{2,n+1}+\Delta t\,\delta G_{2,n}. \end{cases}$$

On the bounded set fixed by M_v and M_p, the maps G_1, G_2 are Lipschitz. Thus there are $L_P>0$ and $C_P^0>0$ such that

$$\|\delta G_{1,n}\|_{\ell^\infty}+\|\delta G_{2,n}\|_{\ell^\infty} \le L_P\left(\|\delta p_{1,n+1}\|_{\ell^\infty}+\|\delta p_{2,n+1}\|_{\ell^\infty}\right)+C_P^0\left(\|\delta v_{1,n}\|_{\ell^\infty}+\|\delta v_{2,n}\|_{\ell^\infty}+\|\delta u_{2,n}\|_{\ell^\infty}\right).$$

Using $\|(A_i^*)^{-1}\|_\infty \le 1$ and

$$E_n^p := \|\delta p_{1,n}\|_{\ell^\infty}+\|\delta p_{2,n}\|_{\ell^\infty},$$

we obtain

$$E_n^p \le (1+\Delta t L_P)E_{n+1}^p+\Delta t\,C_P^0\left(\|v_h-\widehat{v}_h\|_{\infty,h}+\|u_{2,h}-\widehat{u}_{2,h}\|_{\infty,h}\right).$$

Since $E_N^p=0$, backward discrete Gronwall gives

$$\|p_h-\widehat{p}_h\|_{\infty,h} \le C_{P,h}\left(\|v_h-\widehat{v}_h\|_{\infty,h}+\|u_{2,h}-\widehat{u}_{2,h}\|_{\infty,h}\right),$$

where $C_{P,h}=C_P^0Te^{L_PT}$ is admissible. With the state estimate,

$$\|p_h-\widehat{p}_h\|_{\infty,h} \le C_{P,h}(1+C_{S,h})\|u_{2,h}-\widehat{u}_{2,h}\|_{\infty,h}.$$

Step 3: Lipschitz estimate for the control update.

The projection $\Pi_{[0,U_2^{\max}]}$ is 1-Lipschitz. Therefore

$$\|\Phi_h(u_{2,h})-\Phi_h(\widehat{u}_{2,h})\|_{\infty,h} \le (1-\theta)\|u_{2,h}-\widehat{u}_{2,h}\|_{\infty,h}+\theta\frac{q}{\lambda_2}\left\|v_{2,h}(\eta+p_{2,h}^+)-\widehat{v}_{2,h}(\eta+\widehat{p}_{2,h}^+)\right\|_{\infty,h},$$

where

$$p_{2,h}^+ := (p_{2,n+1})_{0\le n\le N-1}, \qquad \widehat{p}_{2,h}^+ := (\widehat{p}_{2,n+1})_{0\le n\le N-1}.$$

Split the product as

$$v_{2,h}(\eta+p_{2,h}^+)-\widehat{v}_{2,h}(\eta+\widehat{p}_{2,h}^+) = (v_{2,h}-\widehat{v}_{2,h})(\eta+p_{2,h}^+)+\widehat{v}_{2,h}(p_{2,h}^+-\widehat{p}_{2,h}^+).$$

The bounds $\|v_{2,h}\|_{\infty,h}\le M_v$ and $\|p_{2,h}^+\|_{\infty,h}\le M_p$ give

$$\left\|v_{2,h}(\eta+p_{2,h}^+)-\widehat{v}_{2,h}(\eta+\widehat{p}_{2,h}^+)\right\|_{\infty,h} \le (\eta+M_p)\|v_{2,h}-\widehat{v}_{2,h}\|_{\infty,h}+M_v\|p_{2,h}^+-\widehat{p}_{2,h}^+\|_{\infty,h}.$$

Moreover,

$$\|p_{2,h}^+-\widehat{p}_{2,h}^+\|_{\infty,h} \le \|p_h-\widehat{p}_h\|_{\infty,h},$$

so the estimates from Steps 1 and 2 yield

$$\|\Phi_h(u_{2,h})-\Phi_h(\widehat{u}_{2,h})\|_{\infty,h} \le \left[(1-\theta)+\theta\frac{q}{\lambda_2}\left((\eta+M_p)C_{S,h}+M_vC_{P,h}(1+C_{S,h})\right)\right]\|u_{2,h}-\widehat{u}_{2,h}\|_{\infty,h}.$$

By the definition of K_h and κ_h,

$$\|\Phi_h(u_{2,h})-\Phi_h(\widehat{u}_{2,h})\|_{\infty,h} \le \kappa_h\|u_{2,h}-\widehat{u}_{2,h}\|_{\infty,h}, \quad \text{with} \quad \kappa_h := (1-\theta)+\theta K_h.$$

Step 4: conclusion.

The preceding estimate proves that Φ_h is Lipschitz with constant $\kappa_h = (1-\theta) + \theta K_h$. Since $0 < \theta \le 1$, we have $\kappa_h < 1$ exactly when $K_h < 1$. Under this condition, Φ_h maps the complete metric space $\mathscr{U}_{\mathrm{ad},h}$ into itself and is a contraction. Banach's fixed-point theorem gives a unique fixed point $u_{2,h}^{\infty}$. For every initial admissible control, the sequence $u_{2,h}^{(k+1)} = \Phi_h(u_{2,h}^{(k)})$ converges to this point in $\|\cdot\|_{\infty,h}$, with the geometric estimate $\|u_{2,h}^{(k)} - u_{2,h}^{\infty}\|_{\infty,h} \le \kappa_h^k \|u_{2,h}^{(0)} - u_{2,h}^{\infty}\|_{\infty,h}$. The state and adjoint Lipschitz estimates then imply

$$v_{i,h}^{(k)} \longrightarrow v_{i,h}^{\infty} \quad \text{and} \quad p_{i,h}^{(k)} \longrightarrow p_{i,h}^{\infty}, \qquad i = 1, 2.$$

Passing to the limit in the state recursion, the adjoint recursion, and the projected control formula shows that

$$(v_{1,h}^{\infty}, v_{2,h}^{\infty}, p_{1,h}^{\infty}, p_{2,h}^{\infty}, u_{2,h}^{\infty})$$

solves the fully discrete optimality system, which completes the proof. □

Remark. For fixed a priori and Lipschitz constants, K_h contains the ratio q/λ_2. Thus $K_h < 1$ is easier to satisfy when q is small or λ_2 is large. This relation also has a modeling meaning: q is the fishing efficiency and λ_2 is the cost of effort. A parameter regime favorable to the contraction proof may therefore describe weak harvesting or expensive effort, while an aggressive and efficient fishing policy can make the fixed-point coupling stronger.

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